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Beyond the Gold Standard: Towards Industrially Viable Electrodes for Durable Perovskite Solar Cells

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ABSTRACT

Perovskite solar cells have achieved high efficiencies but remain limited by instabilities. For the opaque metal rear electrode, Au is commonly used to achieve stable operation but is impractical for scalable photovoltaics due to cost. We investigate the effect of a 5 nm chromium (Cr) interlayer beneath low-cost metals (Al, Ag, Cu) to inhibit metal interdiffusion. Cr/Al electrodes yield devices with high efficiencies (up to 24.7%) and operational stability comparable to reference devices using Au rear electrodes under heat and light stress, with raw material costs reduced by five orders of magnitude. At the front semi-transparent electrode, through which sunlight will be incident, we identify indium diffusion from indium tin oxide (ITO) as a key degradation mechanism, revealed via depth-resolved mass spectrometry. We demonstrate that replacing ITO with fluorine-doped tin oxide (FTO) substantially improves stability. Combining both improvements, FTO with Cr/Al contacts, produces devices retaining >66% of the initial efficiency after >1000 h ageing at 75 °C under simulated sunlight. These results highlight the critical role of electrode selection on perovskite solar cell durability and provide a practical route toward stable and cost-effective perovskite photovoltaics.

1 | Introduction

Photovoltaic technologies are crucial for the transition to low-carbon-emission electricity to prevent a climate catastrophe [1–3]. To exceed the efficiency limits of the current single-junction silicon technology and enable new applications, there has been a strong interest in new emerging solar absorber materials. The majority of these are thin-film technologies, which use different electrode materials from those established for silicon, such as transparent conducting oxides (TCOs) and evaporated metals, as opposed to screen-printed metal grids in silicon photovoltaics. With thin-film technologies attracting more attention, the importance of thorough knowledge of their electrode materials also increases.

Out of the emerging PV technologies, perovskite solar cells (PSC) and perovskite-silicon-tandem solar cells have rapidly increased in power conversion efficiencies (PCEs) over the last decade, reaching certified values of over 28% and 35%, respectively, as of mid-2026 [4]. A major challenge to translating this laboratory success to industry is the stability and scalability of the perovskite technology. While much of the literature focuses on stabilizing the perovskite light absorber, the stability impact of other materials in the device stack, including the front and rear electrodes, has received little attention.

It has been shown that the rear metal electrode materials have a substantial influence on PSC operational stability [5, 6]. More reactive noble metals, like Ag, have been shown to react with the halides that diffuse from the perovskite layer, and metal ions can leach into the perovskite layer, accelerating perovskite degradation [6–9]. Carbon electrodes, which have often been suggested as a stable alternative, are challenging to integrate with the p-i-n architecture of perovskite solar cells, which is considered more efficient, stable and more relevant for multi-junction photovoltaics [10]. This poses a unique challenge for perovskites, since cost-effective industrial metals appear to be difficult to integrate into devices and still achieve high stability. On the other hand, Au is more stable when used in lead halide perovskite solar cells, especially when a thin layer of Cr is deposited below the Au to inhibit metal diffusion into the perovskite [5, 11–13]. However, it is expected to be too expensive for industrial photovoltaic applications, with a 100 nm thick gold layer costing around 60€/m^2 for the material alone (see Figure 5c) [14, 15]. For comparison, at 10 Euro cents/W, an entire silicon PV module costs around 18€/m^2.

While metals can cause instabilities in PSC, TCOs are often used as a more stable alternative and as barrier layers to prevent metal-induced degradation [7, 16]. However, the stability influence of different TCOs themselves remains inadequately studied, particularly for PSCs. The most widely used TCO in both perovskite and perovskite-silicon-tandem solar cells is indium tin oxide (ITO), or other indium-based TCOs such as hydrogenated indium oxide and indium zinc oxide [17–21]. Even though ITO is known to have low chemical and thermal stability [17, 22, 23], there is no comprehensive literature on how these material properties translate to device stability. Zhan et al. found a chemical reaction between ITO/SnO_2 that accelerates the degradation of negative-intrinsic-positive (n-i-p) triple cation PSC because the ITO reacts with and consumes perovskite decomposition products [24]. Furthermore, it has been reported that the use of fluorine-doped tin oxide (FTO), instead of ITO, improves the operational stability of perovskite solar cells [25]. However, there is currently neither a thorough study on full devices confirming that this effect is reproducibly observable, nor sufficient research on the underlying reason for these observed effects.

In this work, we present a comprehensive stability study using positive-intrinsic-negative (p-i-n) formamidinium caesium lead iodide bromide perovskite solar cells, comparing otherwise identical device stacks with different opaque rear- and semi-transparent front-electrodes.

We find that the cheaper industrial metals Cu and Al result in even faster degradation than Ag when used as the electrodes. Encouragingly, we find that a nano-layer of Cr can be used universally beneath all metal electrodes investigated, including the more reactive noble metals like Ag, Cu, and Al, to greatly enhance the stability. The Cr layer strongly improves the stability of all three metals, with Cr/Al even exceeding the stability of Cr/Au electrodes. Using x-ray photoelectron spectroscopy (XPS), we investigated the interfacial chemistry and found that the Cr layer promotes the formation of passivating species, helping to preserve metallic Al under stress. We further demonstrate that at the front of the substrate, changing from ITO to FTO considerably improves device stability. We link the ITO instability to indium species diffusing into the perovskite layer. By evaluating electrode longevity, cost, and sustainability, we identify suitable contacts for scalable thin-film perovskite PV modules using materials compatible with multi-terawatt-scale photovoltaic production [14]. Our findings not only address key challenges in PSCs but also have broader implications for the stability of electrode materials in halogen-rich environments under light and heat exposure and may have implications in other optoelectronic devices, such as LEDs and radiation detectors.

2 | Results

2.1 | Back Electrode—Stability of Cr/Ag Electrodes

A commonly employed back electrode for perovskite solar cells is Ag. As such, we first investigate how Ag changes with a Cr interlayer, which previously has been proven to prevent gold migration. We fabricated otherwise identical p-i-n devices of a layer stack of FTO/poly-TPD/$FA_{0.83}Cs_{0.17}Pb_{1.00}I_{2.7}Br_{0.3}$/PCBM/BCP/Metal (see Figure 1a and Methods for details). Selection of this layer stack was motivated by its documented long-term stability and its consistent performance across different substrates and back-electrode materials [26].

To establish a baseline for intrinsic thermal robustness, we began by characterizing device stability during elevated temperature ageing in a dark nitrogen environment, isolating thermal effects in the absence of oxygen and light. We compared devices with either a 100 nm Ag electrode, a Cr/Ag bilayer, or a Cr/Au bilayer (5 and 100 nm, respectively), aged at 85 °C in a nitrogen atmosphere without encapsulation (ISOS D1) for 20,000 h. Ag and Cr/Au were included as established unstable and stable references, respectively, while Cr/Ag represents the electrode architecture explored further in this work. This is one of the longest reported accelerated ageing studies of PSC. Remarkably, after 20,000 h of thermal ageing, the champion device with a Cr/Au electrode maintained 85% of its maximum performance, which represents <1% average relative degradation per 1000 h. The Cr/Ag device maintained ~50% of its maximum performance, and the device with the neat Ag electrodes only retained 30% of its maximum efficiency (Figure 1e).

A major reason for the higher stability of Cr/Ag devices, as compared to neat Ag, was the sustained open-circuit voltage (V_{OC}) with all devices of this category still outputting over 1 V after 20,000 h, while all Ag electrode devices had a V_{OC} of ~0.8 V (Figure 1c).

Both Cr/Ag and Ag devices showed a strong decrease in short-circuit current density (J_{SC}), falling to median values of 11.0 and

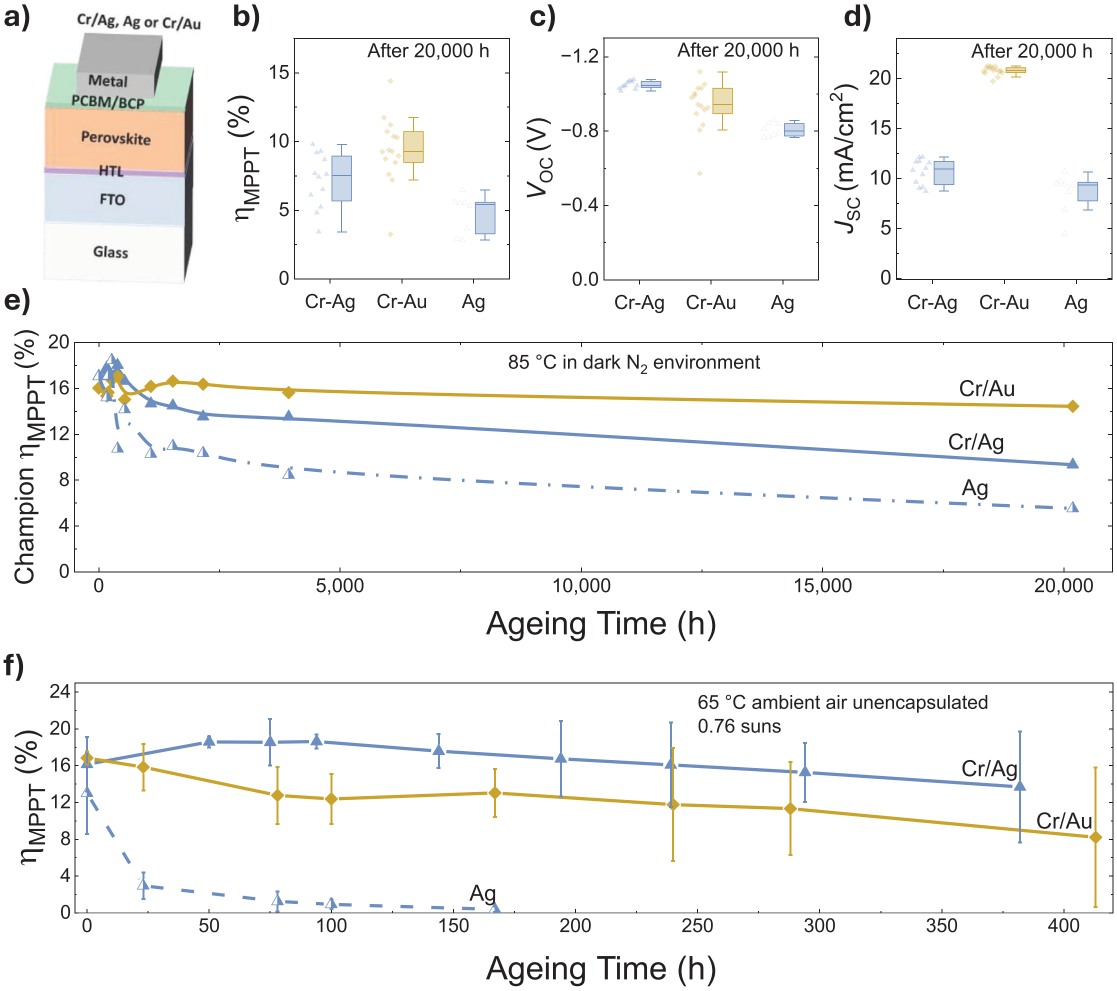

FIGURE 1 | Initial Cr/Ag stability results (a) The device stack used in the experiments. (b)—(d) show the statistics of MPPT efficiency, stabilized V_{OC}, and stabilized J_{SC}, respectively, of all devices of each group with Cr/Ag, Cr/Au, and Ag electrodes after being aged at 85 °C in a dark nitrogen environment for 20,000 h. Boxes represent the 25%–75% interquartile range (IQR), the lines across boxes represent the medians, the whiskers are 1.5 IQR, and filled symbols represent individual datapoints. Each data point is an individual device measured under an AAA solar simulator. (e) MPPT efficiency performance of the best of the three categories during the 20,000 h of ageing. (f) Maximum power point tracked (MPPT) efficiency of devices with a Cr/Ag, Ag, or Cr/Au electrode over time when aged without any encapsulation at 65 °C under 0.76 suns in ambient air at open-circuit conditions. The cells were removed from the ageing box at the measurement time intervals, allowed to cool to room temperature, and measured under a AAA solar simulator. The error bars indicate the average deviation from the median, and the symbols and line represent the median MPPT efficiency. Non-functioning devices were excluded, and statistics are based on 9 Cr/Ag (3 excluded), 13 Ag (3 excluded), and 14 Cr/Au (2 excluded) devices.

9.3 mA/cm^2, respectively, while Au devices maintained a median J_{SC} of 20.9 mA/cm^2 (Figure 1d). The low J_{SC} but high V_{OC} of Cr/Ag devices can be explained by the fact that the devices were aged unencapsulated. While the Cr protected the buried interface, the top of the Ag could still react with halides effused from the periphery of the device active area, which is consistent with the tarnished optical appearance of the electrode after ageing (Figures S1 and S2).

Temperature alone, as used in this first stability study, is not a very strong stressor for perovskite solar cells. Exposing cells to simulated sunlight at an elevated temperature, analogous to the ISOS-L2 stress test, is generally more challenging [27, 28]. So, we exposed our unencapsulated PSCs with Cr/Ag, Ag, and Cr/Au electrodes to 0.76 sun, in an ageing box with a xenon lamp and no UV filter (open-circuit conditions) in air at 65 °C, as set by a black temperature standard (see Methods) (Figure 1f). The best Cr/Ag device had an initial maximum power point tracked efficiency of 18.0% (increasing to 19.2% during ageing) and retained 97% of its maximum performance after 380 h under these relatively harsh ageing conditions. The Cr/Ag devices maintained a higher median MPPT efficiency than Cr/Au devices, whereas Ag devices dropped to a median MPPT efficiency of less than 3% in 24 h (Figure 1f).

2.2 | Back Electrode—General Stability of Cr Metal Bilayers

To explore if the Cr nano-layer generally enhances the stability of PSCs when employed with more reactive metals, we extended the investigation from Ag to also include Cu and Al electrodes.

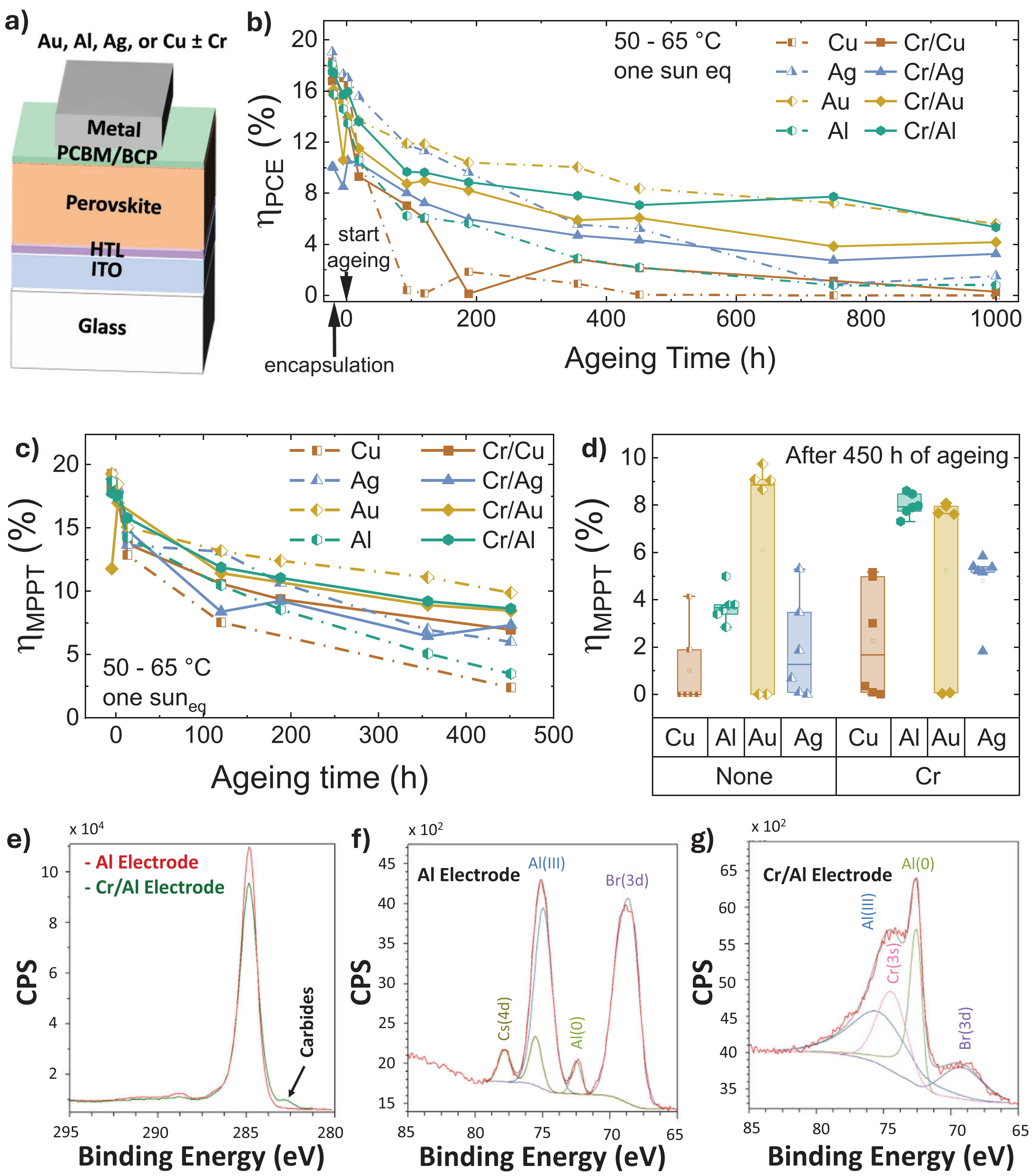

FIGURE 2 | Cr/metal ageing study. (a) The device stack used. (b) The median *JV*-derived PCE of all devices in each electrode category as a function of ageing time of devices completed with an Ag, Au, Al, and Cu electrode with and without a Cr layer. The PCEs were obtained via *JV*-sweeps in an AAA solar simulator. All devices were aged under one sun-equivalent illumination, at elevated temperatures of 59 ± 9 °C, and at open-circuit conditions. The lines indicate the median of all PCEs of both sweep directions of 18 devices per parameter. Yellow rhomboids, green octagons, blue triangles, and orange squares display Au, Al, Ag, and Cu electrode data. Electrodes without chrome are symbolized by half-filled symbols and dashed lines, and electrodes with a chrome interlayer are displayed with a filled symbol and continuous lines. In addition to the population statistics shown in (b), one representative high-performing device from each electrode category was selected for MPPT measurements. The resulting operational stability data are shown in (c). After 450 h of ageing, a 30 s MPPT efficiency was performed on all six devices on the best-performing substrate. The statistical distribution of these measurements is shown in (d). Because of a measurement error, the MPP data were not acquired beyond 450 h of ageing. (e) The C(1s) spectra of an aged Al and a Cr/Al electrode. (f) and (g) the spectra of an Al and a Cr/Al electrode, respectively, buried interface after 1,500 h of ageing at 75 ± 5 °C at 0.76 suns (open circuit conditions).

These two metals were chosen due to their cost-effectiveness, abundance, and excellent conductivity. Devices, identical in all other aspects except their back electrodes, were fabricated using Au, Ag, Al, and Cu, with and without a Cr interlayer. The same device structure as used in the Cr/Ag vs. Ag study was used, with the only difference being a change from FTO to ITO to obtain slightly higher performance because of the higher transparency of the latter (Figure 2a). In pristine devices, there was no major difference in device performance for any of the different metal electrodes, all performing with PCEs of around 20% (Figures S5 and S6).

Devices with a Cr buffer layer, however, showed a wider spread in performance parameters and a slightly increased hysteresis

TABLE 1 | PCE and J_{SC} ageing data. Columns 2 and 3 show the median PCE out of all scans in both directions and the champion PCE out of the single best scan right after fabrication and after ageing for 1000 h under 1.0 suns at 59 ± 9 °C (temperature measured on back encapsulation glass, aged at open circuit conditions in ambient air under a Sulfur lamp).

Electrode	Median (Champion) PCE 0 h	Median (Champion) PCE after 1000 h	Champion MPPT 0 h	Champion MPPT after 450 h	Median *Jsc* initial	Median *Jsc* after 1000 h
Materials	[%]	[%]	[%]	[%]	[mA/cm^2]	[mA/cm^2]
Al	18.1 (19.4)	0.8 (3.2)	18.7	3.5	21.8	4.9
Cr/Al	16 (20.5)	5.3 (8.8)	17.8	8.6	23.2	12.2
Au	17.5 (19)	4.2 (9.8)	18.5	9.9	21.9	13.2
Cr/Au	18.1 (20.1)	5.6 (8.2)	17.0	8.5	21.7	10.1
Ag	19.5 (20.4)	1.5 (5.4)	19.1	6.0	22.5	3.8
Cr/Ag	10.8 (18.6)	3.3 (6.4)	17.9	7.3	21.6	9.5
Cu	18.3 (20.4)	0.0 (2.9)	19.3	2.4	22.2	0.0
Cr/Cu	17.3 (20.2)	0.3 (6.1)	18.3	7.0	22.3	3.2

compared to neat metal counterparts (Figure S5). For Cr/Ag devices, the number of lower-performing devices was more than half, so the median PCE fell within the cluster of low-performing devices at ~10% (Figure S5 and Table 1). It is frequently discussed in the literature that the work function of the chosen electrode material influences device performance (choosing deeper work function electrodes like Au for n-i-p and shallower ones like Cu for p-i-n) [29, 30]. We, however, did not observe any trend, since all tested metals showed very similar open-circuit voltages and device performance (Figures S5 and S6). This indicates that metals can be used interchangeably as electrodes for PSCs almost independently of their work function in vacuum, presumably due to good charge selectivity being achieved with the charge transport layers.

Devices with a neat metal layer showed more frequent “connection errors” during measurement than devices with the Cr interlayer. In approximately 27% of the devices without Cr, failure occurred, most of them connection errors, leading to inconsistent or non-representative measurements. Only 15% of the devices with the Cr interlayer were “non-functional”, and most of the non-functional cells were “shunted”. However, we observed a higher fraction of non-functional cells in devices with Cr layers that were deposited via thermal evaporation from a chrome-coated tungsten wire, with 38 out of 120 devices (32%) being non-functional. In contrast, we observed a much higher yield with all devices (i.e., 64 out of 64 devices) being functional when we deposited the Cr layer via e-beam evaporation (see Figure S8). We speculate in the SI as to the origin of this Cr deposition issue, but further work is required to fully understand this.

To be able to compare devices with and without Cr, all devices were encapsulated with a glass coverslip using a UV-cured epoxy glue and then aged at elevated temperatures above 59 ± 9 °C, under one sun-equivalent sulfur lamp (open-circuit conditions), and in ambient air. In Figure 2b we show the median PCE stability of all scans in both scan directions as a function of ageing time. In addition to the PCE statistics for all device groups, we also measured the MPPT efficiency (60s scans) for one champion device of each group for the first 450 h of ageing, confirming the same trend that we observed for the median PCE data as described below (Figure 2c,d).

For devices with more reactive metals, specifically Ag, Cu, and Al, we observed a substantially slowed-down degradation in device performance when a Cr layer was deposited between the ETL and the metal (see Figure 2b,c and Figures S7 and S10). Cu and Al devices without a Cr layer showed very rapid initial degradation in MPPT efficiency and PCE, with faster degradation than the Ag-contacted devices. Especially Cu-electrode devices exhibited the largest spread in ageing behavior, which leads to apparent dips in the median PCE trace at individual measurement points (Figure 2b); after 1000 h, more than half of the Cu devices were non-operational, giving median PCE and J_{SC} values of 0.0, although a small number of devices still retained measurable performance (Table 1). The time it took for the champion Cu device to lose 50% of its initial efficiency (T_{50}), i.e., to drop from around 20% to below 10% efficiency, was 157 h. For the Al devices, the T_{50} was 325 h (Figure S10). Conversely, the Cr/Cu and Cr/Al champions exhibited a T_{50} of 440 h and 882 h, respectively. The same trend is confirmed by MPPT efficiency values (Figure 2c,d), with the addition of the Cr layer approximately doubling the longevity of devices with Cu and Al electrodes. In this second study, the effect of the Cr layer was smaller for Ag devices, leading only to a small improvement after long ageing times of 1000 h. Overall, the median and champion PCE and J_{SC} values before and after 1000 h of ageing clearly show a strong stability enhancement induced by the Cr layer for all metals but Au (Table 1). While a back electrode with just Al is the second most rapidly degrading option in our stability comparison, the Cr/Al combination shows the second highest stability, very similar to the stability of devices with neat Au electrodes and notably higher than the ones with Cr/Au electrodes, under the ageing conditions in this study (Figure 2a,b).

All device types showed only a slight loss in voltage output during ageing, with a slow, uniform decline in median V_{OC}, from about 1.10 V to roughly 0.95 V (Figure S10b). The only exceptions were devices with Cu and Cr/Cu electrodes, which showed a stronger decrease in median V_{OC}, from 1.10 V to around 0.8 V and 0.9 V, respectively. The slower degradation in devices with a 5 nm Cr layer below the metal electrodes was mainly due to better retention of J_{SC} (Figure S10d and Table 1). This trend is also shown in the change in *JV* behavior displayed in Figure S7. Cu devices without Cr dropped to a median J_{SC} of 0 mA/cm^2 after around 450 h of ageing, explaining their low performance. While the Cr/Cu devices maintained around 12 mA/cm^2 after 450 h, their median J_{SC} also dropped significantly to less than 4 mA/cm^2 after 1000 h (Figure S10d and Table 1). After 1000 h, Al devices decreased in J_{SC} to a median value of 4.9 mA/cm^2, while Cr/Al devices decreased to 12.2 mA/cm^2. A similar trend was visible with a median J_{SC} of around 3.8 mA/cm^2 for Ag and 9.5 mA/cm^2 for Cr/Ag devices after 1000 h (Figure S10a,c).

Since the main difference in stability between devices with and without a 5 nm Cr layer is a slower reduction in current density (see Figure S10d), we postulate damage to the perovskite layer itself to be a major part of the degradation of devices without a Cr layer. More reactive metals than Au are more likely to react with halides that are liberated from the perovskite layer and form metal halides themselves, consistent with the lower electrochemical nobility of Ag, Cu, and Al compared with Au (Table S6). This provides a sink for the degradation product of the perovskite layer, shifting the reaction equilibrium and accelerating the decomposition reaction [31]. If a thin metal halide layer forms on the surface of the metal and electron transport layer, this would reduce conductivity and shift the energetics at that interface. Such changes could explain the faster reduction in the fill factor of devices without the Cr layer.

We postulate that the Cr layer inhibits the reaction between the halides leaving the perovskite and the metal back electrode. This effect slows down the perovskite degradation significantly and is visible across all three more reactive metals tested. We assume, therefore, that Cr can be used as a universal stability-enhancing interlayer beneath metals prone to reaction with halides. We observed a less reproducible effect with Cr/Au electrodes, consistent with the fact that Au is less reactive to halides than the other metals in our study.

To elucidate the chemical difference between electrodes with and without a Cr interlayer, we conducted x-ray photoelectron spectroscopy. The XPS analysis was focused on Al and Cr/Al because Al showed the fastest degradation, whereas Cr/Al was the most stable Cr-containing electrode. Comparing these limiting cases provides a sensitive assessment of electrode-driven degradation and the extent to which Cr suppresses it. The improved stability of Ag and Cu devices with Cr interlayers indicates a similar stabilising effect, although the interfacial chemistry may differ between metals. We probed the buried interface between the ETL and the metal electrode in multiple spots on Al and Cr/Al devices aged for 1500 h at 75 ± 5 °C and 0.76 suns (see Methods and Supporting Information for more details). One difference between the two electrodes was noted in the C(1s) spectra (Figure 2e). For the Cr/Al-electrode, an extra feature was present as a shoulder on the low binding energy side of the C(1s) spectra at 283.9 eV. This feature was unexpected. The equivalent carbide peak is not present on any C(1s) spectra for the Al electrode.

In all cases, the metallic Al electrodes are visible through the carbon films (remnant ETL material). High-resolution Al(2p) spectra were recorded for both Cr/Al and Al samples, and additional Cr(2p) spectra were observed only for the Cr/Al electrodes (Figure 2f). While the aluminium signal in the aged Cr/Al electrodes was mainly of the Al(0) peak at 72.5 eV, the aged Al electrodes consistently showed only a small Al(0) component and, additionally, a large and broad peak assigned to Al(III) at 74.9 eV (Figure 2g) [32].

For the detected mean quantities of C, O, N, and I, there was no significant difference between the spectra of Al and Cr/Al samples (Table S3). However, the signal of the metal electrode was much stronger for the Cr/Al than for the Al samples. On the other hand, the mean Pb signal was 0.20 at.% for the Al and 0.04 at.% for the Cr/Al spectra. For the Cr/Al samples, we consistently detected no or very low Br and Cs (Figure 2f and Table S3). However, the Al electrode samples showed a mean Br signal of 0.7 at.% and a Cs mean signal of 0.03 at.%. The mean Br signal in Al spectra was 0.7 at.%, almost double the detected I signal of 0.4 at.%.

Because of the large excess of Br relative to Pb and Cs and the high signal of Al(III) that was present in all Al (but none of the Cr/Al) spectra, we assume the formation of $AlBr_3$ or other aluminum bromide species to be a major degradation product in aluminum electrodes without the Cr layer. Given that the pristine perovskite contained 90% I and only 10% Br, the disproportionately high Br signal relative to I suggests a degradation pathway selectively driven by the minority halide. Such a preferential Br-induced degradation in mixed-halide perovskites appears to be previously unreported and may have been overlooked in prior studies. Another possible explanation might be a selective loss of iodide species via the formation of volatile iodine gas during the XPS measurement [33, 34].

The second noticeable difference between Al and Cr/Al XPS is the formation of a shoulder in the C1(s) signal. We assign this shoulder to carbon-containing Cr species, consistent with possible chromium carbide formation [35]. Chromium carbides have been reported to have high hardness, wear resistance [36, 37], and corrosion resistance, including to acidic halide solutions [38, 39]. Cr-C displays a strong covalent bond, especially in the most common Cr_2C_3 carbide [39, 40]. We therefore infer that the Cr-containing interlayer may include carbon-containing Cr species in addition to metallic and oxidized Cr.

The combined evidence supports the presence of a mixed Cr/CrOx interlayer after deposition and ageing. In the Cr/Al electrodes, the Cr(2p) spectra show a dominant Cr(0) contribution together with a higher-binding-energy shoulder, indicating that part of the Cr is involved in additional chemical bonding, consistent with oxidized Cr species (Figure S49). This interpretation is further supported by literature reports of oxidized components in vacuum-deposited Cr films and by the green coloration of the Cr deposition source, which is consistent with chromium oxide formation [11]. We therefore propose that the Cr interlayer acts as a mixed metallic/oxidized and possibly carbon-containing barrier layer, suppressing halide transport to the reactive Al electrode and thereby reducing Al halide formation.

TABLE 2 | T80 and T70 times. The times in hours for the median maximum power point tracked efficiency of each device group to drop below 80% (T80) and 70% (T70) of its maximum median efficiency. For the FTO/Cr/Ag device group, the median efficiency did not fall below 70% within the 400 h duration of Study 1.

Study	Transparent Electrode	Opaque Electrode	T_{80} [h]	T_{70} [h]
1	FTO	Cr/Au	66	240
1	FTO	Cr/Ag	315	>400
2	ITO	Cr/Au	28	66
2	ITO	Cr/Ag	80	115

Both chromium and aluminum oxides are known for their passivating behavior towards acids and halides, and Al_2O_3 is widely used as a stable passivation layer in silicon solar cells [41–45]. However, the XPS results indicate that substantial halide-driven reactions nevertheless occur in neat Al electrodes, resulting in the formation of Al(III) species and halide-containing degradation products. Given the high reactivity of Al towards halides, we propose that any native oxide present is insufficient to prevent aluminum halide formation under device operating conditions, explaining the poor stability of Al electrodes without a Cr interlayer. On the other side, it has been reported that the thermal and chemical stability of $(Al_{1-x}Cr_x)_2O_3$ is much higher than that of Cr_2O_3 [46], with Al suppressing the formation of higher-valent Cr species [46–48]. Further, the carbon, which we infer to be chromium carbide, detected in XPS spectra of Cr/Al electrodes can form in a solid-state exothermic thermite reaction between Cr_2O_3, Al, and carbon (see SI) [49, 50]. Such mixed oxide formation, together with the carbon signal detected in the Cr/Al XPS spectra, could contribute to the formation of a chemically robust buried interlayer that limits halide-driven degradation at the Al electrode.

2.3 | Front electrode—Stability of Transparent Conductive Oxides

Returning to our solar cell stability studies, we observed a lower overall stability in the devices shown in Figure 2 compared to the stability presented in Figure 1. Whereas devices in the first study (Figure 1f) showed a gradual decrease, devices in the second study (Figure 2) showed a strong exponential decay in efficiency over time. This caused, for example, devices with Cr/Au electrodes in the second study to fall below their median T_{80} within only 28 h, while devices within the same group in the first study fell below their median T_{80} after 66 h, more than double that time (see Table 2). The T_{80} for Cr/Ag devices reduced from 315 h in the first study to 80 h in the second.

Since the transparent electrode used was a major difference between the two studies (FTO in Figure 1 and ITO for devices in Figure 2), we next investigated the stability of FTO as compared to ITO-based devices, employing the most stable electrode in the second study (Au). Perovskite solar cells with the same device stack on ITO and FTO substrates were aged under elevated temperature and illumination (open-circuit conditions) and underwent in-depth post-mortem characterisation.

Both ITO- and FTO-based devices show a decay of their steady-state performance measured via maximum power point tracking (MPPT efficiency), shown in Figure 3a. The ITO champion before the start of the ageing experiment started with a steady-state efficiency of 18.1%, which is 1% higher than that of the highest-performing FTO device at that time, with 17.1%. After 800 h of ageing, all 14 ITO devices in the ageing study performed with < 0.1% absolute efficiency, while the FTO device, on average, retained 2.4%, with the most stable FTO device retaining 7.0% MPPT efficiency. The evolution of different ageing parameters, as well as representative current-density-voltage (*JV*) curves, is shown in Figure S12. ITO-based devices showed a fast drop in all performance parameters within the first 400 h for the majority of devices.

A small number of ITO-based devices showed a non-representative high stability initially, but changed ageing behavior to a much quicker degradation after about 200 h. Most ITO-based devices developed "s-kink" current-voltage scan shapes, whereas the devices on FTO substrates did not (Figure S12).

To understand why the ITO samples degraded much faster, time-of-flight secondary ion mass spectroscopy (TOF-SIMS) was performed on an as-fabricated device and on an ITO-based device aged for 800 h under 0.76 sun illumination at 65 °C with an identical layer stack. We observed indium ions throughout the device stack in the pristine, well-performing ITO device as well as in the aged device (Figures S13 and S14). In the fresh device, surprisingly, the indium showed large concentrations peaking in the middle of the perovskite layer and to a lesser extent at the perovskite ETL interface (opposite side to the ITO electrode). In contrast, for the aged devices, the indium distribution monotonically decreased going from the ITO electrode towards the ETL layer. An accumulation of indium cations in the top surface of perovskite films in aged ITO-based devices was observed by Zhan et al. [24] In n-i-p devices, they observed a more homogeneous distribution of indium species in fresh devices and an accumulation at the HTL/perovskite interface in aged devices. Because of the indium species being present in our as-fabricated devices, we conclude that significant indium leaching happens already during fabrication, likely during the annealing steps at 100 °C. Some indium ions may also be leached out via cationic exchange from the perovskite precursor salts during the film deposition and casting process.

Scanning electron microscopy (SEM) cross-sections of aged devices on ITO and FTO show a different morphology of the perovskite layer. ITO-based devices, as shown in Figure 3d and Figure S15, show several tens of nanometres in size cavities, especially near the ITO interface, as well as around grain boundaries throughout the whole film thickness. These cavities are not visible in devices aged on FTO substrates (Figure 3e and Figure S16) or in pristine ITO-based devices (Figure S52).

To investigate the isolated effect of the ITO substrate on the perovskite material under the ageing conditions, we aged perovskite films identical to the active layer in our solar cell devices on

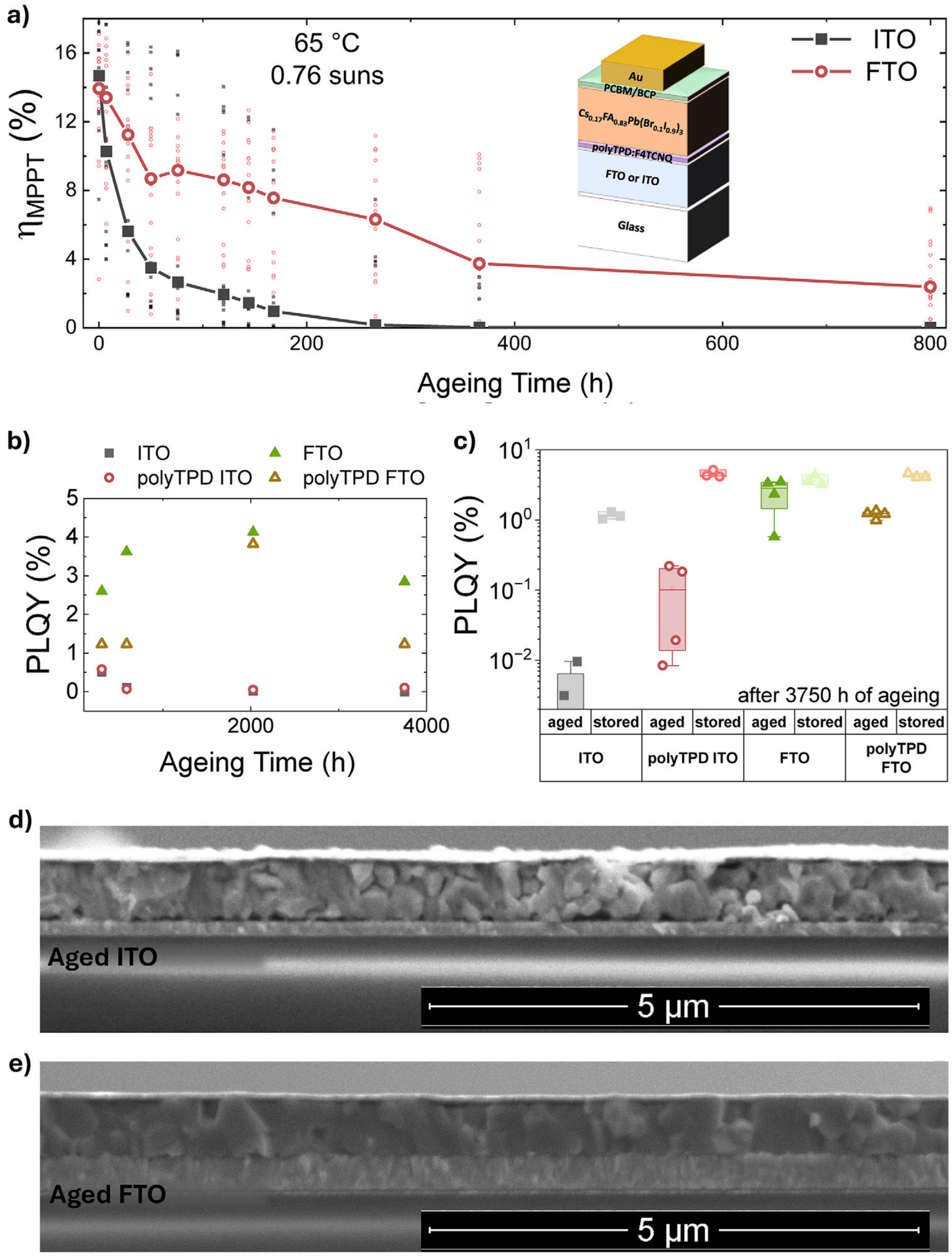

FIGURE 3 | Stability of devices on ITO and FTO. (a) shows the average MPPT efficiency performance of full PSC devices on FTO (in red line and circles) and ITO substrates (black line and squares) when the devices were aged under 65 °C and 0.76 sun illumination in ambient air (open circuit conditions). The data of all individual devices of each category are shown with the same symbol. The device stack is shown in the inset. All devices were completed with an Au electrode. The ageing behavior of photoluminescence quantum yield (PLQY) of perovskite film on ITO and FTO is shown in (b) and (c). Perovskite films were aged under 0.76 sun equivalent illumination and at temperatures between 68 ± 3 °C in a nitrogen atmosphere for 3750 h (open circuit conditions). The perovskite films were deposited either directly on FTO or ITO substrates or as half-stacks on top of the HTL poly-TPD on top of ITO or FTO. On these films, PLQY and TRPL were periodically measured during the ageing. The development of the PLQY is shown in (b), and statistics of the PLQY of aged films and films stored in the dark at room temperature in nitrogen are displayed in (c). SEM cross-sections of ITO devices and FTO devices after being aged under 0.76 sun equivalent illumination at room temperature for 800 h are shown in (d) and (e), respectively. Additional cross-sections of aged and fresh ITO and FTO devices can be found in Figures S15–S18 and S52–S53.

different substrates. The substrates included glass, ITO, ITO/poly-TPD, FTO, and FTO/poly-TPD. The ageing was conducted at temperatures of 68 ± 3 °C under a three color LED array with close to one sun equivalent irradiance in a nitrogen atmosphere for 3750 h. Reference samples were kept in the dark in a nitrogen atmosphere at room temperature. Time-resolved photoluminescence (TRPL) and photoluminescence quantum yield (PLQY) measurements were conducted periodically throughout the ageing period to monitor the quality of the perovskite layer.

The PLQY results are shown in Figure 3b,c (for full results see Figure S19). The PL signal of films on an ITO or ITO/poly-TPD substrate rapidly dropped from above 1 to below 0.01% upon ageing (Figure 3b). In contrast, perovskite on FTO or

FTO/poly-TPD substrates kept PLQY values between 1 and 5% throughout the entirety of the ageing time. All reference devices kept at room temperature, besides the ones on uncoated ITO, kept PLQY values of above 3%. Interestingly, the stored ITO sample exhibited a lower PLQY (~1%) than the other stored reference films (>3%), even without deliberate light or heat stress.

We postulate that this drop in PLQY is due to indium cations leaching out of the ITO substrate. The drop in PLQY observed in ITO/perovskite samples stored in the dark suggests that this occurs even in the absence of light or heat, and is primarily driven by a solid-state interaction at the ITO-perovskite interface. Heat and light exposure may accelerate this diffusion. For half-stacks with different layers between the perovskite and ITO, such as polyTPD or Poly(methyl methacrylate) (PMMA), the PLQY still decreases under light and heat. This indicates that the layers between the perovskite and ITO, such as polyTPD or PMMA, can slow but not fully prevent In^{3+} leaching interaction. This may be due to two reasons: First, achieving a perfectly continuous layer of polyTPD or PMMA between the perovskite and ITO is difficult for typically nanometre-thick layers. Any discontinuity would create direct contact between the perovskite and ITO, acting as a starting point for the ion leaching. Second, even if a fully continuous layer could be achieved, In^{3+} and halide ions can likely diffuse through it under light and heat exposure, allowing the reaction to continue. If a high enough content of In^{3+} ions diffuses into the perovskite, they are expected to create deep trap states, which will act as non-radiative recombination centres, reducing PLQY. These results are consistent with Zhou et al. [51], who studied the effect of indium doping on methylammonium lead tri-iodide perovskites. They showed that replacing as much as 1% of the lead stoichiometry with indium could negatively influence the photoluminescence of the resulting perovskite films. The authors further showed that In^{3+} doping into $MAPbI_3$ causes mid-band-gap states to form, which in turn is consistent with reduced PL.

To distinguish between interface- and bulk-driven degradation pathways, time-resolved photoluminescence (TRPL) measurements were performed throughout the ageing study on the same half-stacks using two excitation wavelengths. When excited with a 405 nm laser from the TCO/HTL/perovskite side, the fitted lifetimes of FTO-based and glass reference samples remained broadly stable, with both bare FTO and glass ending at approximately 15 ns after 3750 h of ageing. In contrast, ITO-based samples showed a catastrophic lifetime loss, with bare ITO and ITO/polyTPD decreasing from between 10 and 100 ns to below 1 ns (Figure S21a). A similar trend was observed when a 635 nm excitation laser was used from the substrate/perovskite side. Bare ITO decreased from approximately 300 ns to below 2 ns, while ITO/polyTPD decreased from approximately 200 ns to below 5 ns after 3750 h of ageing (Figure S21b). FTO-based samples, in contrast, remained highly stable, with bare FTO even increasing from approximately 280 ns to approximately 490 ns (close to the glass reference) over 3750 h of ageing. TRPL decays from perovskite films deposited on a glass slide were stable throughout the ageing, independently of the excitation laser used.

The different probing depths of the two excitation wavelengths provide insight into the spatial origin of the half-stack degradation. Excitation at 405 nm is predominantly absorbed close to the illuminated substrate/perovskite side, whereas 635 nm excitation penetrates much deeper into the perovskite layer. The pronounced lifetime decrease in ITO-based samples under both excitation wavelengths indicates that degradation is not limited to a single interfacial region. Instead, the strong reduction in lifetime observed for ITO-based samples under 635 nm excitation indicates substantial degradation within the bulk of the perovskite layer (Figure S21b). This observation is consistent with the TOF-SIMS results showing indium distributed throughout the perovskite film and supports the conclusion that indium leaching from ITO induces bulk defect formation rather than merely degrading the immediate TCO/HTL/perovskite interface.

ITO is known to exhibit strong instability in acidic environments, undergoing degradation through acid-driven etching reactions [22, 23, 52]. In the context of perovskite solar cells, this instability is particularly relevant due to the presence of weak Brønsted acid salts, such as formamidinium-based compounds. Additionally, metal species within the device stack, including diffused rear metal and metallic lead or caesium inside the perovskite, may act as catalysts to etching reactions, potentially accelerating ITO degradation and hence release of indium, over time. This highlights the susceptibility of ITO to chemical instability in PSC devices in particular [53].

Kerner et al. [53]. showed that when ITO films are in direct contact with a methylammonium lead triiodide ($MAPbI_3$) or a formamidinium lead triiodide ($FAPbI_3$) perovskite layer, a solid-state reaction between the materials can be triggered upon exposure to temperatures of 85 °C or −0.8 V of applied bias. Kerner et al. were able to detect 1 at.% of InI_3 on the top surface of a 200 nm thick $MAPbI_3$ film via x-ray photo spectroscopy (XPS) when aged on top of an ITO substrate at 100 °C for 1 h. Further, they showed irreversible damage to an ITO/$MAPbI_3$/Au device when cycled once to -0.8 V [53]. via cyclic voltammetry. This also allows to reasonably argue for a voltage-induced degradation reaction of the ITO layer under operational conditions.

This shows that relatively large amounts of In^{3+} have the potential to be leached out of the ITO and diffuse throughout the perovskite layer. These results also align well with our analysis of TOF-SIMS on aged and pristine perovskite solar cell devices (see Figures S13 and S14). While quantitative analysis of the concentration of indium inside the perovskite layer is difficult, we estimate the indium content relative to lead to be in the order of 1% and 0.1% in pristine and aged ITO-based devices. We estimate that only 0.17% and 0.51% of the indium contained in a 100 nm thick ITO layer needs to leach out to reach 1% and 3% indium concentration relative to lead in the 500 nm thick perovskite layer (see calculation in Supporting Information).

Based on our TOF-SIMS results, we investigated the influence of intentionally adding indium ions to the perovskite layer by fabricating solar cell devices and thin films with a range of added indium iodide concentrations from 0.001% to 2% molar excess (relative to lead halide salts added to the precursor solution) on FTO substrates. Upon fabrication, there was no significant difference in initial performance between no added indium iodide (0%) and small amounts of added indium iodide (0.001% – 0.1%), with 0.1% even showing slightly improved efficiency (Figure 4a). However, we observed a reduced PLQY for indium

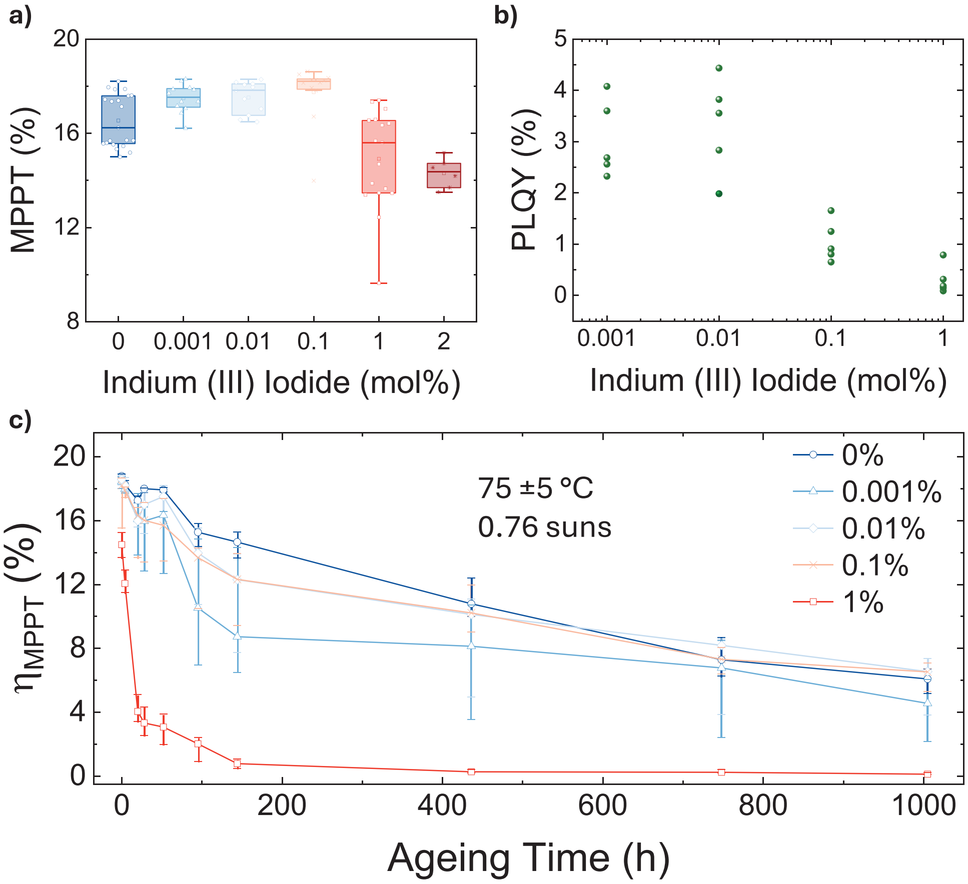

FIGURE 4 | Influence of added Indium (III) Iodide. (a) The stabilized performance of devices with different percentages of excess Indium (III) Iodide added into the bulk perovskite of FTO-based devices. For devices with 1% and 2% added indium, the performance drops, while it increases slightly for smaller amounts. (b) The Photoluminescence Quantum Yield (PLQY) for films with the same amount of added Indium (III). There are signs of a reduction in PLQY for 0.1% of indium excess in perovskite films. The ageing behavior of the same devices depicted in (a) is shown in (c) for an ageing condition of 0.76 suns, 75 ± 5 °C (open circuit conditions), encapsulated, and in ambient air. For each condition, 12 devices were aged. The solid line with symbols represents the median for each condition, while the whiskers of the corresponding color denote the interquartile range (first to third quartile). Devices with 1% added indium show a strongly accelerated decrease in efficiency, while smaller amounts show a behavior similar to devices without added indium.

iodide concentrations as low as 0.1% (Figure 4b).

For large amounts of added indium excess (1%–2%), there was a sharp drop in median performance of the perovskite solar cells from around 18% down to around 14%, largely due to a voltage drop (see Figure S24b,d). To probe the effect of indium inside the perovskite layer on device stability, we aged these devices at 75 ± 5°C, under 0.76 sun illumination, and at open-circuit conditions. The devices with large amounts of indium (1%) degraded to 4% median performance within 20 h of ageing (Figure 4c). The ageing behavior of the devices on FTO substrates with 1% added indium was very similar to the ITO-based devices with no indium iodide, showing the same characteristics in a strong initial decrease of current density and development of "s-kinks" (compare Figures S12d and S51b). On the other hand, lower indium iodide concentrations maintained more than 15% median efficiency for the first day of ageing. The smaller indium concentrations showed a comparable rate of degradation to devices without added indium (0%). A 0.001% of excess indium iodide showed the second-highest degradation rate, while 0.01 and 0.1% aged almost identically to 0% (Figure 4c). However, the differences between the 0%, 0.001%, 0.01%, and 0.1% indium concentrations, as compared to the 1% concentrations, are relatively small considering the device-to-device variation observed within each group. We thus do not consider the slightly faster degradation of the 0.001% group to represent a systematic concentration-dependent effect (see Figure S25). In contrast, the 1% indium devices exhibit a clear and reproducible acceleration of degradation.

We only observed a negative effect of indium ions on device stability for the larger concentration of 1% indium iodide. While this seems relatively large for impurities, our TOF-SIMS results, as well as results by Kerner et al. [53] indicate indium ions are in fact present in the perovskite films processed on ITO substrates at this order of magnitude. Our results indicate that the indium in the perovskite layer at the start of the ageing triggers an accelerated photo-induced degradation of the perovskite material in the device. Because we determine that the highest concentration of indium content in the perovskite films is in fresh devices, we assume that most indium leaches into the perovskite during fabrication, 65 °C ageing temperature does not appear to cause significant further indium leaching, and the leaching reaction is not specifically triggered by light.

Beyond the presence of indium, there are two relevant differences that we can consider between the ITO and FTO substrates. The first is the surface roughness, and the second is the type of glass used, adhesion and blocking layers between the glass and the TCO. The latter may influence the possibility of other ions leaching out of the substrate into the perovskite film, specifically sodium. Concerning surface roughness, mechanical issues could contribute to the degradation of PSC devices, and these mechanical issues may be more pronounced on the smoother ITO substrates than on the rougher FTO substrates. These explanations are, however, unlikely to be the main factors since we would not expect delamination at this interface to result in a reduced PLQY of perovskite films aged on ITO substrates. Further, the degradation in charge carrier lifetimes when measuring TRPL using a 635 nm laser cannot be explained by delamination. We expect to probe the bulk perovskite layer with this measurement based on the expected penetration depth of a laser with this wavelength. The faster drop in PL decay lifetime for perovskite films processed on ITO substrates can only be explained by a degradation of the bulk perovskite layer and not simply by degradation of the interface. The cavities in the SEM cross-sections could be explained by mechanical delamination, for example, due to thermal cycling during the measurement of the devices. Another possible explanation could be the perovskite layer being much richer in In^{3+} ions at the bottom interface, which in turn could accelerate the degradation and material loss near that location, resulting in cavity formation.

In our TOF-SIMS results (see Figures S13 and S14), we also detect significant concentrations of sodium ions in the perovskite layer, with concentrations exceeding those of In. It has been suggested that Na^+ could substitute for lead in the perovskite structure, resulting in a negative Na'_{Pb} defect [54]. This defect has, however, been proposed to have a positive instead of a negative effect on the optoelectronic properties of the perovskite, increasing conductivity and stability to heat- and light-induced degradation [54–57]. Further, as shown in Figures S19 and S21, films were more stable on a microscope glass slide than on any other substrate. These slides are made of soda lime glass and contain 13–16 w/w % Na_2O [58]. If Na^+ doping were the dominant degradation pathway under these conditions, we would expect films on this glass to be unstable.

Finally, our observation that the PSCs fabricated on FTO substrates but with ≥ 1% indium doping degraded much faster than the devices containing lower or no added indium, is most simply explained by the presence of indium in the perovskite being the root-cause of the accelerated degradation.

The diverse range of ageing conditions and electrode materials investigated in this study reveals several important insights (Table 3). Firstly, the stability of Cr/Au-based devices under different ageing conditions shows notable differences. While the champion device maintains a T_{85} of 20,000 h at 85 °C in the dark, its lifetime is dramatically reduced to about 90 h under simulated 0.76 sun illumination at a similar elevated temperature (Figure 1b and Figure 5). This demonstrates the critical role of light exposure at elevated temperatures in degradation pathways, highlighting the need for combined heat and light stability testing.

2.4 | Combining Both Electrodes—Perovskite Devices With Improved Long-Term Stability

Because of the instability we have revealed when using ITO electrodes, we repeated the metal electrode comparison with the two most stable electrodes, Au and Cr/Al, on FTO instead of ITO substrates. This final study was performed in a different laboratory (Oxford), to also test lab-to-lab reproducibility and at an increased ageing temperature of 75 ± 5 °C under 0.76 suns (as opposed to 59 ± 9 °C for the previous study). ITO devices with Cr/Au electrodes were added for a repeated comparison of the stability influence of the transparent electrodes. The Cr/Al devices again showed strong stability (Figure 5a and Figure S26), even exceeding the stability of devices with Au electrodes (FTO devices with Au electrodes are the 0% references of Figure 4). Interestingly, we were not able to fabricate working devices with an Al electrode without any Cr layer in Oxford because the devices degraded during the encapsulation process (see Figure S27).

After 1037 h at 75 ± 5 °C under light, the best-performing Cr/Al device achieved an MPPT efficiency of 11.2% (66% of its initial performance), with a corresponding T_{70} of ~630 h. We also compared different thicknesses of the Cr layer of 2.5, 5.0, and 7.5 nm below the Al. The result confirmed that the chosen thickness of 5 nm is the best in terms of long-term stability, as shown in Figure S29. To further investigate the reproducibility of the findings throughout this paper, we aged alongside the FTO-Au and FTO-Cr/Al devices, also FTO-Cr/Au, ITO-Cr/Al, and ITO-Cr/Au devices at 0.76 suns and 75 ± 5 °C. Both device groups with ITO-based transparent electrodes showed faster degradation, with ITO-Cr/Al devices being notably more stable than ITO-Cr/Au devices. Surprisingly, FTO-Cr/Au devices showed low stability, degrading to lower efficiencies than the ITO-Cr/Al devices after approximately 800 h (Figure S32).

In order to investigate the general applicability of the Cr/Al electrode, we tested the device performance using different stacks together with the FTO and Cr/Al electrodes. Besides the 1.6 eV bandgap perovskite ($FA_{0.83}Cs_{0.17}PbI_{2.7}Br_{0.3}$) used for the presented stability studies, we employed a wider bandgap ($Cs_{0.1}FA_{0.9}PbI_{2.55}Br_{0.45}$) and a 1.55 eV narrower bandgap perovskite ($Cs_{0.05}FA_{0.90}MA_{0.05}PbI_{2.95}Br_{0.05}$). Devices with the 1.55 eV perovskite together with FTO and Cr/Al reached a PCE of 24.7% (MPPT of 23.8%), showing the applicability of this electrode configuration to high-efficiency cells (Figure 5b). Using a wider bandgap perovskite, we achieved MPPT efficiencies of 20.8% (median 20.2%) and 32.5% under AM1.5 100mWcm^{-2} from a AAA solar simulator and under 1000 lux white LED (correlated color temperature of 3000 K) illumination with relevance to indoor PV applications, respectively (Figure S32). While the wider bandgap perovskite devices were facile to fabricate and could be tested in ambient air, we encountered initial issues with the 1.55 eV devices. The perovskite reacted with the Al on top of the Cr on the edges of the electrode, especially upon exposure to oxygen. We addressed this issue by using a slightly larger evaporation mask for Cr than for Al and immediately encapsulating the device (similar to Figure S3c,d).

In addition to the light-heat stability, we compared the reverse-bias stability of devices with Cr/Al and Au electrodes. Firstly,

TABLE 3 | Overview of ageing studies.

Parameter	Light Source	Temp. Estimate	Temperature Control	Atmosphere	Encapsulation	Number of devices per parameter	Figures
Cr vs none Au, Ag, Al, Cu	Sulfur lamp	58 ± 9 °C (on substrate backside)	Fans (and ambient thermometer)	Ambient Air	MoO_X and a glass coverslip with epoxy	12–18	Figure 2, Figures S7, S10, and S11
FTO vs ITO	Xenon lamp	65 °C (black standard temperature)	Fans	Ambient Air	Glass coverslip with edge epoxy	14 ITO, 16 FTO	Figure 3a and Figure S23
FTO vs ITO	LED three-color growing lamp	58 ± 3 °C	Oven with a metal plate	Nitrogen	PMMA	1 substrate	Figures 3b-c, Figures S19 and S21
FTO vs ITO	LED three-color growing lamp	25 °C	Peltier cooler and fans	Ambient Air	Glass coverslip with edge epoxy	15 ITO, 14 FTO	Figure S22
Indium Iodide added	Xenon lamp	75 ± 5°C (black standard temperature)	Fans	Ambient Air	Glass coverslip with full-area epoxy	12–18	Figure 4c and Figures S24 and S25
Cr/Al vs Au ITO vs FTO Cr thickness	Xenon lamp	75 ± 5 °C (black standard temperature)	Fans	Ambient Air	Glass coverslip with full-area epoxy	12–24	Figure 5, Figures S26–31
Cr/Ag vs Cr/Au vs Ag	None	85 °C	Oven with a metal plate	Nitrogen	None	8–16	Figure 1, Figures S1–S5

we explored the resistance of each device type to reverse-bias-induced breakdown (Figure S33a). The same device was sequentially exposed to successive reverse-bias steps for 60 s each, starting with −0.1 V and increasing the applied voltage in 0.1 V increments up to a reverse-bias of −2.7 V. While applying each voltage, we measured the current density output for each reverse bias. While the Au-based device showed signs of breakdown after cumulative reverse-bias exposures up to −1.1 V, the devices with a Cr/Al electrode showed fully blocking behavior and close to zero current density until cumulative exposure of up to −1.7 V (Figure S33b). Further, for all applied reverse biases, we observed lower reverse-bias current densities for the Cr/Al than for the Au electrode devices, consistent with the improved reverse-bias stability of Cr-containing contacts that has also recently been reported for Cr/Cu electrodes in Sn–Pb perovskite solar cells [59]. After each reverse-bias step, forward and reverse *JV* scans were acquired under one sun illumination to probe the device diode behavior and PCE. While the Au-based device started to display strong deviation from the initial *JV* behavior (consistent with the change in current density under reverse bias), Cr/Al devices showed little change until reverse-bias stress of −1.7 V. At −2.7 V, however, the PCEs of Cr/Al fell below the PCEs of the Au-based device (Figure S33c,d). Following a 24 h recovery period in the dark, the Au-based device showed only a small recovery from 5.2% to 7.6%. However, the average PCE for the scans on the Cr/Al increased strongly from 1.9% to 9.4%. When left for a further 72 h recovery (96 h in total), the Cr/Al device recovered even further to an average PCE of 12.2% while the Au-based device plateaued at 7.4% (Figure S34e).

The general loss in performance of both devices showed elements of decreased shunt resistance, current density and voltage. A possible origin of shunt resistance-related performance loss is the formation of metal filaments [60–63]. The lower shunt resistance to metal filaments in devices with a Cr interlayer is, however, recoverable, which we postulate to be due to the filament losing its conductivity easily due to Cr oxidation reactions. Throughout this study, we have observed "fully shunted" devices (FF = 0.25, PCE < 2%) recovering to ∼ 19% PCE (see SI). Another reported pathway of reverse-bias breakdown is related to the accumulation of mobile ions and halide diffusion into transport layers, creating a reverse-bias breakdown [31, 63]. In contrast to the formed Cr filaments, this breakdown is less reversible because it causes more damage to the perovskite and transport layers. This may explain the low level of recovery of voltage, current density, and EQE signal in the Au-based devices compared to Cr/Al-based devices.

3 | Conclusions

In conclusion, we show that a wide range of metals can be used on identical device stacks without major differences in resulting initial device efficiency. However, using cheap industrial metals Cu and Al results in prohibitively low device stability when aged under light at elevated temperatures. Using a Cr layer stabilizing an Al electrode, we present an electrode configuration with comparable stability to Au electrodes when used in PSCs. What is most surprising is that of the metals investigated, Al is the only metal more reactive than Pb; however, combined with the Cr interlayer, it leads to the most stable cells.

Our observation that indium diffuses into the perovskite from ITO under standard processing conditions before stressing has even begun has broad implications. Architectures employing self-assembled monolayer (SAM) hole contacts may be particularly susceptible due to incomplete surface coverage. Mixed Sn–Pb perovskites, processed from more acidic precursor formulations, and high-temperature-processed (up to >180 °C) absorbers such as evaporated films or $CsPbI_3$ are expected to exhibit accelerated ingress. Given ITO's widespread use in perovskite–silicon tandems and as a recombination contact in perovskite-based multijunction, such diffusion pathways could significantly limit stability in these architectures. Mitigation will require transparent materials that avoid indium migration to maintain long-term device performance. We show how the indium-induced degradation can be mitigated by replacing ITO with FTO. Our work should reinvigorate efforts upon developing new indium-free TCOs, but not only for indium scarcity reasons, but due to stability implications. Combining FTO with Cr/Al electrodes yields devices with stability surpassing Au while reducing cost and avoiding scarce indium.

4 | Methods

4.1 | Materials

The dopant 2,3,5,6-Tetrafluoro-7,7,8,8-tetracyanoquinodimethane (F4-TCNQ, CAS No. 29261-33-4) was purchased from Lumtec, and the hole transporter Poly(4-butylphenyl-diphenyl-amine) (poly-TPD, CAS No. 472960-35-3) was purchased from 1-Material. Lead iodide (PbI2, 99.999%, metal base, CAS No. 10101-63-0), caesium iodide (CsI, 99.99%, CAS No. 7789-17-5), and lead bromide (PbBr2, ultra dry, 99.999%, metal basis, CAS No. 10031-22-8) were all purchased from Alfa-Aesar, Thermo Fisher. Formamidinium iodide (FAI, CAS No. 879643-71-7), (2-(4-(bis(4-methoxyphenyl)amino)phenyl)-1-cyanovinyl)phosphonic acid (MPA-CPA) were purchased from Dyenamo AB. [6, 6]-phenyl-C61-butyric acid methyl ester (PC61BM, >99.5%, CAS No. 160848-22-6) was purchased from Solenne BV. Bathocuproine (BCP, 98%, CAS No.4733-39-5) was purchased from Xi'an Polymer Light Technology Corp. Tetrakis(dimethylamino)Tin (TDMASn, CAS No. 1066-77-9) for ALD and AP-CVD precursor was bought from Pegasus Chemicals. Deionised water (7732-18-5) ALD was used as the oxidizer. NiO_x nanoparticles (2.5 wt.%, ethanol) were purchased from Avantama. All other materials and solvents were, unless stated otherwise, purchased from Sigma-Aldrich. All materials were used as received without further treatment or purification.

4.2 | Precursor Solutions

If not mentioned otherwise, all solutions were prepared in a nitrogen-filled glovebox and filtered with a 22 µm PTFE filter. The hole transport layer solution was prepared by dissolving 1 mg/mL poly-TPD with 20 wt.% F4-TCNQ in toluene by stirring overnight at 70 °C–80 °C. NiO_x nanoparticles were diluted in ethanol (v/v, 1:10) by vortexing for 1 min before use. MPA-CPA was dissolved at 1 mg/mL in ethanol at room temperature by vortexing for 1 min before use. The perovskite precursor solution was prepared by stirring overnight 3.3 mg $[BMP]^+$ $[BF4]^-$, 256.2 mg

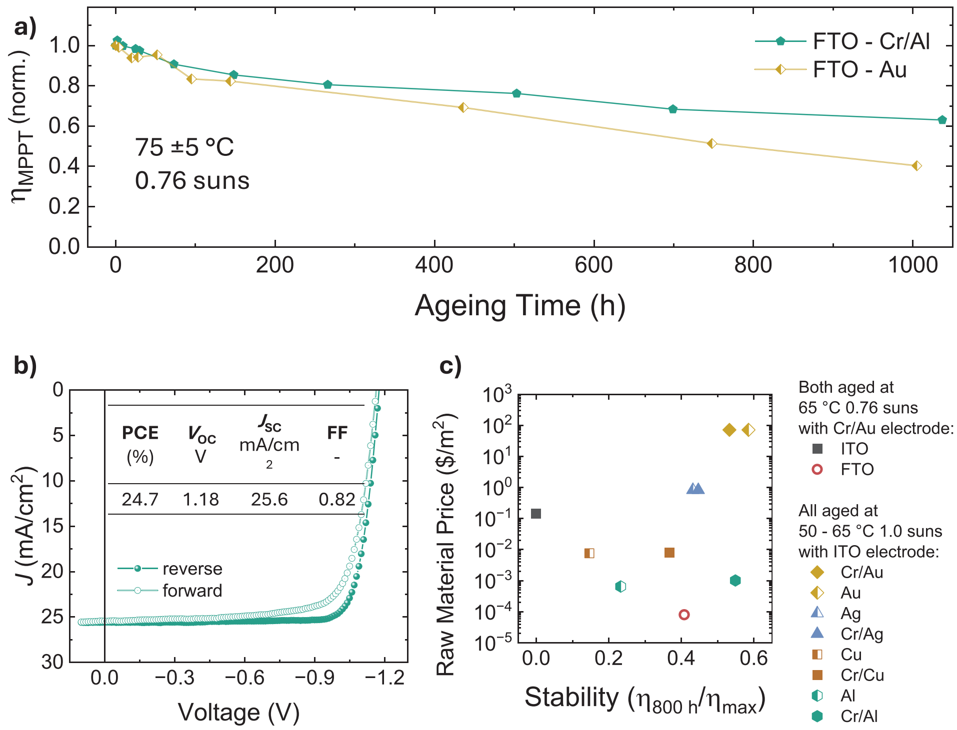

FIGURE 5 | Ageing behavior of Cr/Al and Au devices on FTO and summary stability and price. (a) The normalized ageing behavior of the champion stability FACs device with FTO and Cr/Al electrodes and the champion with FTO and Au electrodes. The efficiencies were obtained via maximum power point tracking measured under a AAA solar simulator, and devices were aged under 0.76 sun-equivalent illumination at 75 ± 5 °C, encapsulated in ambient air, and at open-circuit conditions. The devices were aged in the same ageing box in overlapping intervals. The not-normalized data of the full batches are shown in Figure S26. (b) The forward and reverse current density voltage (*JV*) scans of the high record efficiency CsFAMA perovskite device with Cr/Al device on FTO substrates and the *JV* derived performance parameters. (c) Different opaque and transparent electrode materials are plotted according to the price of the metal raw materials contained in 1 m^2 of the electrodes prepared against the ratio of the maximum median efficiencies against the median efficiency of PSC with these electrodes after 800 h of heat-light ageing. The transparent electrodes, ITO and FTO, were aged with Cr/Au back electrodes in a separate study from the different metal electrodes, which were all aged with ITO transparent electrodes. For the ITO vs. FTO study, the ageing conditions were set to 65 °C and 0.76 suns (see Figure 3), and for the metal study, determined as 59 ± 9 °C at 1.0 suns (see Figure 2 and Table 3). We calculated the metal raw material prices based on market prices and the volume bound in 1 m^2 of an electrode as used in a laboratory-scale device. For ITO and FTO, we only considered the prices of the Sn and In bound in the film as calculated by Wagner et al. [14]. Preparation costs were not included.

CsI, 827.9 mg FAI, 2295.5 mg PbI_2, and 319.2 mg $PbBr_2$ in 3.2 mL DMF and 0.8 mL DMSO to obtain 4 mL of 1.45 M solution of $Cs_{0.17}FA_{0.83}Pb_{1.01}I_{2.73}Br_{0.27}$ in 4:1 DMF:DMSO. The wider band gap perovskite $Cs_{0.1}FA_{0.9}PbI_{2.55}Br_{0.45}$ was prepared by dissolving 39 mg CsI, 123.9 mg $PbBr_2$, 535.9 mg PbI_2, and 232.2 mg FAI in 0.8 mL DMF and 0.2 mL DMSO at room temperature for 1 h. PCBM solution was prepared by dissolving 20 mg/mL in 3:1 di-chlorobenzene:chlorobenzene. BCP was dissolved at a concentration of 1 mg/mL in IPA.

4.3 | Fabrication of Perovskite Solar Cells

FTO substrates were either bought pre-patterned from Latech. ITO substrates were purchased from Biotain or. Devices were scrubbed with a toothbrush with water and Fairy washing up liquid, rinsed with DI water, and sonicated with a mix of DI water with 10% Decon90 cleaning liquid, DI water, acetone, and IPA for 15 min in each solvent. Substrates were then UV ozone-treated for 15 min. Poly-TPD was spin-coated dynamically at 2000 rpm for 20 s and followed by an anneal in ambient air for 7 min at 130 °C. Subsequently, all substrates were transferred into a nitrogen-filled glovebox. The perovskite layer was spin-coated with 250 µL of precursor solution in a two-step programme at 1000 rpm for 5 s (5 s ramp up) and 5000 rpm for 30 s (5 s ramp up). At 40 s total spin time, 400 µL of anisole antisolvent was dispensed on the spinning substrate. The films were immediately transferred onto a hot plate for a 1-h anneal at 100 °C. Substrates were left for a 10-minute cool down on cleanroom wipes, followed by PCBM spin-coating dynamically at 2000 rpm for 20 s and annealed for 5 min at 100 °C. A BCP layer was spun onto the PCBM with a spin-speed of 5000 rpm for 30 s, followed by a one-minute drying step at 100 °C.

For the wider band gap PSCs, NiO_x nanoparticles were spin-coated on cleaned FTO substrates at 3000 rpm for 20 s (3 s ramp up), followed by depositing MPA-CPA at 3000 rpm for 20 s (3 s ramp up) atop and annealing at 100 °C for 10 min. After cooling down, 200 µL $Cs_{0.1}FA_{0.9}PbI_{2.55}Br_{0.45}$ was spin-coated at 1000 rpm for 10 s (2 s ramp up) and 5000 rpm for 30 s (2 s ramp up). 150 µL anisole was cast 5 s before the end. The samples were annealed at 100 °C for 10 min. 20 nm of C_{60} and 140 cycles (~20 nm) of

ALD-SnO_x were sequentially deposited as the electron transport layer.

For the narrow band gap PSCs, the hole-selective layer, NiOx (diluted in 1:10 ethanol, v:v) was spin-coated in the air at 3,000 rpm for 30 s (3 s acceleration to 3,000 rpm) without post-treatment. The hole-selective layer was deposited with Me4PACz (0.5 mg ml−1 in IPA) solution by spin-coating inside an N2-filled glovebox, and the substrates were annealed at 105 °C for 10 min.

To deposit the perovskite film with the main composition of $Cs_{0.05}FA_{0.90}MA_{0.05}PbI_{2.95}Br_{0.05}$, 1.6 M perovskite precursor solution was prepared by mixing CsI, FAI, MABr and PbI2 in DMF:DMSO (4:1, v:v) mixed solvent subject to the stoichiometric ratio [64]. Further, 2 mol% PbI2, 2 mol% PbCl2, and 5 mol% MACl were added to the precursor as additives for better film quality [65]. The perovskite precursor (200 µl) was spin-coated at 4000 rpm for 50 s (5 s acceleration to 4,000 rpm). 270 µl anisole was dropped on the film 20 s before the end of the spinning programme. The film was immediately annealed at 105 °C for 30 min. For the interfacial passivation layer, a solution with 0.5 mg EDAI2 and 0.5 mg PEACl added to 1 ml IPA and chlorobenzene (v:v, 1:1) mixed solvent was spin-coated onto the as-prepared perovskite films and followed up with 3 min of thermal annealing under 105 °C [66–68].

4.4 | Preparation of the Electrodes

We thermally evaporated metal layers at pressures below $2.0{*}10^{-6}$ torr.

4.5 | Solar Simulator Measurements

Devices were measured with a Keithley under an AAA class solar LED array by Wavelabs. An automated programme was used to measure the devices. First stabilized V_{OC} values were obtained by measuring at zero current for 10 s. This was followed by current-voltage sweeps in both directions through the range of 50 to −1400 mV performed at a scan speed of 245 mV/s. Sweeps were stopped if the current density exceeded 50 mA/cm^2. The dark sweeps were conducted first, followed by the sweeps under illumination. Then, a 30 s MPPT efficiency was conducted using a gradient descent algorithm. Finally, a 10 s stabilized J_{SC} tracking at zero bias was conducted. This measurement protocol was for initial performance assessment as well as for all measurements during ageing experiments.

4.6 | Photoluminescence Quantum Yield

The Photoluminescence Quantum Yield (PLQY) was conducted as proposed by De Mello et al. [69], using a 532 nm continuous wave laser excitation source (Roithner, RLTMLL-532 2 W) to excite a sample inside an integrating sphere (Newport, 70682NS), and the laser scatter and PL were collected using a fibre-coupled spectrometer (Ocean Optics QEPro). The beam intensity was modified using a continuous filter to match the absorbed photon flux for the 1.6 eV bandgap under 1 sun AM1.5 illumination (59.6 mW/cm^2) [70]. The laser was illuminated from the glass side for all samples. The background signal was subtracted using an empty sphere in the dark as a reference.

4.7 | Time-Resolved Photoluminescence

The photoluminescence decay curves were collected on a FluoTime 300 (PicoQuant GmbH) using a TimeHarp 260 as a time-correlated single photon counting setup (TCSPC) and a pulsed laser diode with an excitation wavelength of 405 nm (LHD-P-C-405, PicoQuant GmbH; 0.5 MHz, fluence: 4.82×1011 cm^{-2}) and a 635 nm laser (LDH-P-635; 0.2 MHz, fluence: 3.01×1011 cm^{-2}) through the glass substrate. The TRPL decays were fitted using a stretched-exponential function, $I(t) = Aexp[-(t/\tau)^{\beta}] + y_0$, where τ is the characteristic lifetime, and β is the stretch factor. β can assume values between 0 and 1, where 1 gives a monoexponential decay while values smaller than 1 indicate a heterogeneous distribution of lifetimes. Here, we calculate the average lifetime from τ and β as $\langle t \rangle = \frac{\tau}{\beta}\,\Gamma(\frac{1}{\beta})$, with $\Gamma\,(\frac{1}{\beta}) = \int_0^\infty x^{\frac{1-\beta}{\beta}}\, e^{-x}\, dx$. The average lifetimes were used for the comparison of ageing-dependent lifetime evolution. In metal halide perovskite thin films β has recently been related to homogeneity of the PL emission [71].

4.8 | Scanning Electron Microscopy

Samples for cross-sections were prepared by diamond scribing the edges and cleaving the substrates mechanically. The breaking point was set to lie in the active area of devices. The samples were grounded to the sample holder with conductive copper tape. A Hitachi S-4300 scanning electron microscope was used for imaging. A working distance of 8–10 mm, a spot size of 3.0, and an acceleration voltage of 5 kV were used.

4.9 | X-Ray Diffraction

The samples for x-ray diffraction (XRD) were prepared on FTO or ITO substrates as full devices. Patterns were obtained from using a Cu Kα x-ray source ($\lambda = 1.5406$ Å) and a Panalytical X'PERT Pro x-ray diffractometer. Scans were conducted for 2 h per sample in ambient air, and an automatic sample changer was used.

4.10 | Accelerated Ageing Tests

4.10.1 | Encapsulation

Devices were encapsulated with a glass slide glued to the back side of the substrate. UV-activated epoxy (Everlight Eversolar AB-341) was either distributed uniformly over the active area or along the edge of the glass slide. The epoxy was applied to the glass encapsulation slide before the substrate was placed in the correct position onto the epoxy glass slide, followed by 3 min of UV-curing. To ensure good glass-epoxy-substrate contact and avoid edge effects, perovskite material was removed from the substrate at the edge of the encapsulation area. For the devices in the second ageing study, an additional protective layer of MoO_X was evaporated over the device area to protect the device from epoxy damage (Table 3).

4.11 | "Freiburg" Box

Devices were put into a 3D-printed black holder, active side facing up, into a home-built ageing box. The box was illuminated with a sulfur light. The temperature was controlled with varying fan speeds to keep the air temperature in the chamber at 40 °C. The temperature on the back side of the glass encapsulation was measured with a thermocouple Greisinger GTH 1170 Digital Quick Response Thermometer to be between 50 °C and 68 °C (Figure S38). Due to the thermal conductivity of glass, there can be an expected temperature gradient through the encapsulation glass thickness. Hence, the perovskite active layer can be significantly higher than the 50 °C and 68 °C range.

4.11.1 | Atlas Boxes (65 °C and 75 °C Light Ageing)

Devices were placed in a black (anodized) metal rack facing the glass side up in a closed Atlas box Suntest CPS+. The temperature that we targeted for the active layer in the solar cells was 65 °C or 85 °C, which we measured and controlled with a black temperature standard, supplied with the aging box (Table 3). However, to more closely estimate our cell temperature, we also prepared thermocouples bonded to black anodized aluminum foil (Thorlabs BKF12), which had been adhered to an ITO-coated glass slide, with the same format as used for the solar cell devices. The "mock device thermocouples" were mounted in the same location as the solar cell devices in recorded temperatures of 75 ± 5 °C, while the commercially supplied black temperature standard recorded 85 °C. We hence moderate down our estimation of our cell temperature in the ageing boxes to be 75 ± 5 °C when the box is set to nominal 85 °C, and this is what we have stated and reported here in this paper. We note that in our previous publications where we stated 85 °C ageing, the temperature was recorded on the Atlas-supplied black temperature standard, and hence likely to be 75 ± 5 °C, rather than 85 °C. We have no estimate for the real cell temperature of 65 °C ageing, but we assume it is (due to cell reflection) likely slightly lower than 65 °C. The ambient temperature in the chamber was around 35 °C. They were illuminated with a Xenon lamp SunTest CPS Plus simulated AM1.5 sunlight (760 W/m^2). The spectrum of the box is shown in Figure S39. The temperature was regulated by reducing the heat induced by the illumination with varying fan speeds.

The mock devices were prepared by glueing the black anodized aluminum foil (Thorlabs BKF12) to the TCO side of an ITO-coated glass slide using Araldite 24 h two-part epoxy. The epoxy was degassed under vacuum for 10 min. For compression, a vacuum bag was used to squeeze out any extra epoxy from between the TCO glass and aluminum foil. A K-type thermocouple (RS Components part no. 621–2158) was glued to the back side using a thermally conductive epoxy (MG Chemicals 8329TFS). The entire mock device was vacuum bagged to ensure good thermal contact, and the epoxy was cured according to the manufacturer's specifications.

4.12 | Nitrogen Heat-Light Ageing

Perovskite films were placed standing upright in a standard heating element convection oven. The substrates were held by an aluminium plate facing the glass door of the oven, with another aluminum plate above the substrates to prevent direct infrared irradiance of the cell from the heating elements. A thermocouple mounted into the middle of one of the metal plates recorded the temperature. A three-color growing lamp was placed to illuminate the substrates through the oven door. The oven was controlled to be at 65 °C for the first 300 h and then to 85 °C from 300 to 3750 h. The oven, however, could only reach up to 70 °C when set to 85 °C. The whole setup was based inside a nitrogen-filled glovebox. For the Cr/Ag vs Ag nitrogen study, the same oven was operated without the LED light and with an aluminum cover to keep the devices completely in the dark. The dark experiment was run for 20,000 h.

4.13 | Light Ambient Ageing

As shown in Figure S23b, devices were placed glass side up onto a Peltier cooler below a three-color plant growing lamp. The devices were kept at 25 °C. The setup was in ambient air in an air-conditioned laboratory.

4.14 | Nitrogen Heat Ageing

Devices were placed in a rack covered with a metal tray inside an oven kept at 85 °C inside a nitrogen-filled glove box. An aluminum plate was placed above the substrates to prevent direct infrared irradiance of the cell from the heating elements. A thermocouple mounted into the middle of one of the metal plates recorded the temperature.

4.15 | Reverse-Bias Stability Test

To investigate the impact of reverse-bias stress on device performance, a stepwise testing procedure was conducted, using a programmable source meter (Cicci Research). For each selected reverse-bias voltage, the dark current density–voltage (*J*–*V*) characteristic was measured with the sample covered in a black box, sweeping from +1.0 V down to the target reverse-bias voltage. The device was then held at the specified reverse-bias voltage for 60 s, during which the reverse current was continuously monitored to assess current stability under stress. Immediately after the biasing period (approximately 5 s), the device was exposed to standard 1-sun (AM1.5G, using a spectral mismatch calibrated G2V Optics Sunbrick Solar Simulator) illumination, and illuminated *JV* measurements were carried out to evaluate the effect of reverse biasing on the various parameters extracted from the *JV* characteristics. *JV* measurements were conducted without a shadow mask, and the active area of the device was 0.3087 cm^2. This procedure was repeated for each reverse-bias voltage, starting from −0.1 V and incrementing in steps of −0.2–−2.7 V.

4.16 | Indoor Light Testing

A customized LightBox XL (185 mm x 85 mm x 85 mm) fabricated by Lightricity Ltd. was employed as the experimental enclosure. The box is constructed from a black acrylic-like photopolymer,

TABLE 4 | Material Properties Electrodes.

Material	Density (g/cm3)	Density (kg/cm3)	Density (kg/m3)	Thickness (nm)
Au	1.93E + 01	1.93E-02	1.93E + 04	100
Ag	1.05E + 01	1.05E-02	1.05E + 04	100
ITO (In)	5.47E + 00	5.47E-03	5.47E + 03	100
ITO (Sn)	1.57E-01	1.57E-04	1.57E + 02	100
FTO (Sn)	5.65E-01	5.65E-04	5.65E + 02	500
Cu	8.94E + 00	8.94E-03	8.94E + 03	100
Al	2.70E + 00	2.70E-03	2.70E + 03	100
Cr	7.16E + 00	7.16E-03	7.16E + 03	5
perovskite	1.08E + 03	1.08E + 00	1.08E + 06	500

which minimizes internal reflections and external light interference. The integrated light source consists of a diffuse 11×11 LED array with a correlated color temperature (CCT) of 3000 K (for the Spectrum see Figure S40). It is mounted 60 mm above the device under test (DUT), with incident light oriented normal to the DUT surface. Calibration of the light source was carried out using an EXL1 1V20 photovoltaic cell and a Keithley 2400 source meter. The total uncertainty associated with the calibration process is below 4%, while the uncertainty in the light intensity measurement remains under 1%. During measurements, a scanning rate of 0.03 V/s was applied. Light intensities were varied across 1000, 500, 200, 100, and 50 lux; a diffuser was introduced for intensities below 500 lux to ensure uniform illumination across the DUT surface.

4.17 | Calculation of Material Costs

The material costs per 1 m^2 P_{m2} of an electrode were calculated from the kg materials prices P_{kg} multiplied by the materials density ρ and the thickness of the layer d typically employed in PSC devices:

$$P_{m2} = P_{kg} \cdot \rho \cdot d$$

The densities of all metals were obtained from the NASA database [72]. For the density of SnO_2, we used values from CRC Handbook of Chemistry and Physics [73]. The density of ITO was calculated as in Wagner et al. [14]. The preparation costs were not included. Assumed thickness for Cr 5 nm, for all other metals 100 nm, for ITO 100 nm, and for FTO 500 nm (Table 4). The raw material prices per kg were calculated from the average price between the values reported by the Federal Institute for Geosciences and Natural Resources (German: Bundesanstalt für Geowissenschaften und Rohstoffe, abbreviated BGR) [15] and the United States Geological Survey USGS [74, 75]. All prices were compared to spot prices reported on Bloomberg.com to check the plausibility of the values (Table 5) [76].

4.18 | X-Ray Photoelectron Spectroscopy (XPS)

All samples were fabricated via mechanical fracture (cleaving) of devices in a nitrogen glove box. In order to examine the buried electron transport layer (ETL) to electrode interface, the back encapsulation glass was mechanically cleaved off the TCO glass substrate, causing a failure inside the ETL. This way, most of the perovskite and ETL would stick to the TCO substrate while the metal electrode would stick to the epoxy on the back encapsulation glass, exposing the ETL/electrode interface (Figure 6).

The fractured devices were immediately placed into the bespoke Kratos glove box sample holder (Kratos Analytical, Manchester, UK) without breaking the nitrogen cycle. The electrode surfaces of interest were affixed to a thick cover glass substrate, which meant that samples were electronically floated and, once placed in the holder, were immediately sealed under nitrogen. The sample holder was transferred immediately (<5 min) into the insertion lock of a Kratos Axis Supra and remained sealed until the insertion lock reached <3.0×10-7 Torr.

XPS analysis was carried out in the Kratos Axis Supra. As samples were floated, the internal charge neutraliser was used throughout the measurement. Samples were analysed without further preparation, except for a short depth profile as noted in the text. When a depth profile was performed, samples were etched in 10s increments with a 3×3 mm rastered 5 kV Ar_{500}^{+} cluster beam. Wide spectra were collected with a "medium" resolution pass energy of 80 eV, 100 ms dwell time and step size of 0.25 eV using a monochromated 225 W Al Kα source and three sweeps from 1200 to −5 eV. Although slower than a traditional wide scan, this allows some chemical shifting to be observed, whilst being fast enough for many regions to be examined. Additionally, 8 regions were examined with more traditional high-resolution spectroscopy to confirm the medium resolution data. High-resolution scans were run with a 40 eV pass energy, 0.1 eV step size, 500 ms dwell and up to 5 sweeps depending on S:N.

Spectra were analyzed on CasaXPS Version 2.3.24PR1.0 using the Kratos sensitivity factor library. As samples were charge neutralized, charge correction was needed. Due to uncertainty around the expected chemical state of the carbon present, all data were charge corrected to the Al(0) peak, which was taken as 72.5 eV. This did result in the primary C(1s) peak being observed at 285 eV. All peak positions were compared to the NIST XPS database in addition to any other citations noted in the main body of the text [77, 78]. Baselines for high-

TABLE 5 | Material Prices.

Element	Price (BGR) ($/kg)	Price (USGS) ($/kg)	Price (spot) ($/kg)	Average ($/kg)
Au	69.381.21	4,311.42	74,927.30	49539.98
Ag	786.09	788.98	942.66	839.2433
In	283.14	240	—	261.57
Sn	27.43	26.15	—	26.79
Cu	8.675	8.4	9.90	8.991333
Al	2.22	2.62	2.57	2.470667
Cr	9.442	11.11	—	10.276

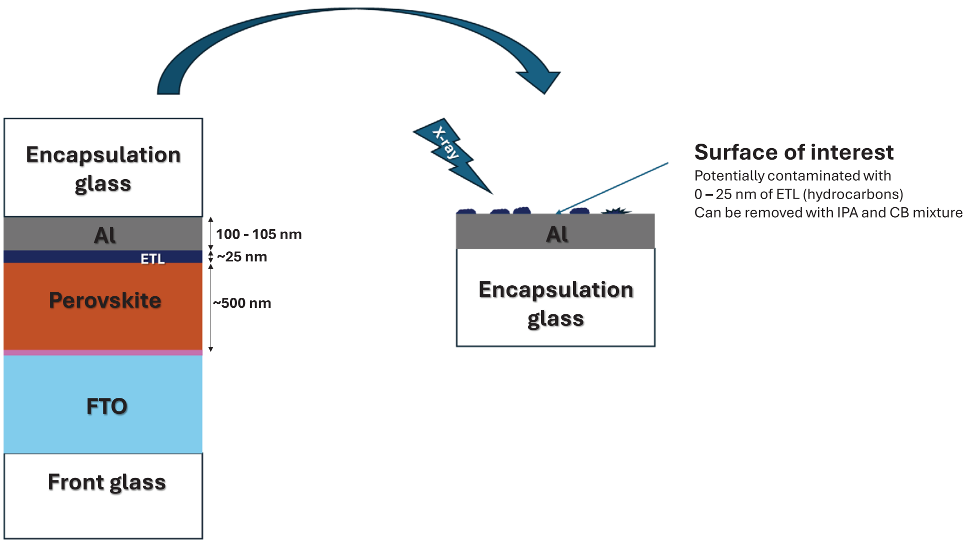

FIGURE 6 | Sketch of x-ray Photoelectron Spectroscopy (XPS) sample preparation.

resolution spectroscopy were Shirley by preference, but in some cases, the complex baselines showed inverted gradients, and a linear baseline was used. Where components were fitted, they were fitted with symmetrical components – aside from metallic peaks in oxidation state zero, where asymmetry is expected. In these cases, peak shapes were as seen previously in the literature [79, 80].

Author Contributions

Tino Lukas: conceptualization, investigation, writing – original draft, methodology, validation, visualization, writing – review and editing, software, formal analysis, data curation, funding acquisition. **Georgios Loukeris**: investigation. **Chia–Yu Chang**: investigation. **Clemens Baretzky**: investigation. **Shuaifeng Hu**: investigation. **James Mcgettrick**: conceptualization, investigation, writing – review and editing, visualization, validation, methodology, data curation, formal analysis. **Manuel Kober-czerny**: data curation, writing – review and editing, methodology, validation, investigation. **Junke Wang**: investigation, writing – review and editing. **Robert L. Z. Hoye**: supervision, resources, writing – review and editing. **Trystan M. Watson**: supervision, funding acquisition. **Markus Kohlstädt**: supervision, project administration, investigation, writing – review and editing, validation, formal analysis. **Ali Reza Nazari Pour**: investigation, visualization, methodology, validation, formal analysis. **Sam Teale**: validation. **Bowei Li**: resources. **Lukas Wagner**: supervision, validation, writing – review and editing. **Martin Stolterfoht**: supervision, resources. **Philippe Holzhey**: conceptualization, validation, writing – review and editing, supervision. **Henry J. Snaith**: conceptualization, supervision, resources, project administration, writing – review and editing, validation, funding acquisition, methodology.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data for this manuscript will be available via the University of Oxford Research Archive [https://ora.ox.ac.uk/], and/or on request from the corresponding author.

References

1. K. Calvin, D. Dasgupta, G. Krinner, et al., *IPCC, 2023: Climate Change*, ed. H. Lee and J. Romero (IPCC, 2023), https://doi.org/10.59327/IPCC/AR6-9789291691647.

2. International Energy Agency, "An Updated Roadmap to Net Zero Emissions by 2050," World Energy Outlook 2022 (Paris: IEA, 2022), https://www.iea.org/reports/world-energy-outlook-2022/an-updated-roadmap-to-net-zero-emissions-by-2050.

3. International Energy Agency, World Energy Outlook 2020 (Paris: IEA, 2020), https://www.iea.org/reports/world-energy-outlook-2020.

4. Best Research-Cell Efficiency Chart | Photovoltaic Research | NREL.

5. K. Domanski, J.-P. Correa-Baena, N. Mine, et al., "Not All That Glitters Is Gold: Metal-Migration-Induced Degradation in Perovskite Solar Cells," *ACS Nano* 10 (2016): 6306–6314, https://doi.org/10.1021/acsnano.6b02613.

6. Y. Kato, L. K. Ono, M. V. Lee, S. Wang, S. R. Raga, and Y. Qi, "Silver Iodide Formation in Methyl Ammonium Lead Iodide Perovskite Solar Cells With Silver Top Electrodes," *Advanced Materials Interfaces* 2, no. 13 (2015): 1500195, https://doi.org/10.1002/admi.201500195.

7. C. C. Boyd, R. Cheacharoen, K. A. Bush, R. Prasanna, T. Leijtens, and M. D. McGehee, "Barrier Design to Prevent Metal-Induced Degradation and Improve Thermal Stability in Perovskite Solar Cells," *ACS Energy Letters* 3, no. 7 (2018): 1772–1778, https://doi.org/10.1021/acsenergylett.8b00926.

8. S. Sirotinskaya, R. Schmechel, and N. Benson, "Influence of the Cathode Microstructure on the Stability of Inverted Planar Perovskite Solar Cells," *RSC Advances* 10, no. 40 (2020): 23653–23661, https://doi.org/10.1039/D0RA00195C.

9. Y. Deng, Q. Dong, C. Bi, Y. Yuan, and J. Huang, "Air-Stable, Efficient Mixed-Cation Perovskite Solar Cells With Cu Electrode by Scalable Fabrication of Active Layer," *Advanced Energy Materials* 6, no. 11 (2016): 1600372, https://doi.org/10.1002/aenm.201600372.

10. T. Lukas, S. Seo, P. Holzhey, et al., "Charge Extraction Multilayers Enable Positive-Intrinsic-Negative Perovskite Solar Cells With Carbon Electrodes," *ACS Energy Letters* 10, no. 6 (2025): 2736–2742, https://doi.org/10.1021/acsenergylett.4c03403.

11. M. Kaltenbrunner, G. Adam, E. D. Glowacki, et al., "Flexible High Power-Per-Weight Perovskite Solar Cells With Chromium Oxide–Metal Contacts for Improved Stability in Air," *Nature Materials* 14, no. 10 (2015): 1032–1039, https://doi.org/10.1038/nmat4388.

12. L. Zhao, R. A. Kerner, Z. Xiao, et al., "Redox Chemistry Dominates the Degradation and Decomposition of Metal Halide Perovskite Optoelectronic Devices," *ACS Energy Letters* 1, no. 3 (2016): 595–602, https://doi.org/10.1021/acsenergylett.6b00320.

13. R. A. Kerner, P. Schulz, J. A. Christians, et al., "Reactions at Noble Metal Contacts With Methylammonium Lead Triiodide Perovskites: Role of underpotential deposition and electrochemistry," *APL Materials* 7, no. 4 (2019): 041103, https://doi.org/10.1063/1.5083812.

14. L. Wagner, J. Suo, B. Yang, et al., "The Resource Demands of Multi-Terawatt-Scale Perovskite Tandem Photovoltaics," *Joule* 8, no. 4 (2024): 1142–1160, https://doi.org/10.1016/j.joule.2024.01.024.

15. Deutsche Rohstoffagentur (DERA), Preismonitor, März 2024 (Berlin: DERA, 2024), https://archive.org/details/dera-2024-preismonitor-maerz-2024.

16. C. C. Boyd, R. Cheacharoen, T. Leijtens, and M. D. McGehee, "Understanding Degradation Mechanisms and Improving Stability of Perovskite Photovoltaics," *Chemical Reviews* 119, no. 5 (2019): 3418–3451, https://doi.org/10.1021/acs.chemrev.8b00336.

17. B. Nath, P. C. Ramamurthy, G. Hegde, and D. R. Mahapatra, "Role of Electrodes on Perovskite Solar Cells Performance: A Review," *ISSS Journal of Micro and Smart Systems* 11, no. 1 (2022): 61–79, https://doi.org/10.1007/s41683-021-00089-y.

18. V. Sittinger, H. King, A. Kaiser, et al., "Indium-Based Transparent Conductive Oxides Developed For Perovskite And Perovskite-Silicon Tandem Solar Cell Applications," *Surface and Coatings Technology* 457 (2023): 129286, https://doi.org/10.1016/j.surfcoat.2023.129286.

19. H. Zhu, B. Shao, Z. Shen, et al., "In Situ Energetics Modulation Enables High-Efficiency and Stable Inverted Perovskite Solar Cells," *Nature Photonics* 19, no. 1 (2025): 28–35, https://doi.org/10.1038/s41566-024-01542-8.

20. F. Hou, X. Ren, H. Guo, et al., "Monolithic Perovskite/Silicon Tandem Solar Cells: A Review of the Present Status and Solutions Toward Commercial Application," *Nano Energy* 124 (2024): 109476, https://doi.org/10.1016/j.nanoen.2024.109476.

21. S. Mariotti, E. Köhnen, F. Scheler, et al., "Interface Engineering for High-Performance, Triple-Halide Perovskite–Silicon Tandem Solar Cells," *Science* 381, no. 6653 (2023): 63–69, https://doi.org/10.1126/science.adf5872.

22. R. A. Afre, N. Sharma, M. Sharon, and M. Sharon, "Transparent Conducting Oxide Films for Various Applications: A Review," *Reviews On Advanced Materials Science* 53, no. 1 (2018): 79–89, https://doi.org/10.1515/rams-2018-0006.

23. J.-H. Choi, S.-O. Kim, D. L. Hilton, and N.-J. Cho, "Acid-Catalyzed Kinetics of Indium Tin Oxide Etching," *Thin Solid Films* 565 (2014): 179–185, https://doi.org/10.1016/j.tsf.2014.06.053.

24. C. Zhan, C. Luo, F. Gao, X. Wang, Y. Li, and Q. Zhao, "Indium Tin Oxide Induced Internal Positive Feedback and Indium Ion Transport in Perovskite Solar Cells," *Angewandte Chemie International Edition* 63, no. 30 (2024): 202403824, https://doi.org/10.1002/anie.202403824.

25. S. M. Park, M. Wei, J. Xu, et al., "Engineering Ligand Reactivity Enables High-Temperature Operation of Stable Perovskite Solar Cells," *Science* 381, no. 6654 (2023): 209–215, https://doi.org/10.1126/science.adi4107.

26. Y.-H. Lin, N. Sakai, P. Da, et al., "A Piperidinium Salt Stabilizes Efficient Metal-Halide Perovskite Solar Cells," *Science* 369, no. 6499 (2020): 96–102, https://doi.org/10.1126/science.aba1628.

27. M. V. Khenkin, E. A. Katz, A. Abate, et al., "Consensus Statement for Stability Assessment and Reporting for Perovskite Photovoltaics Based on ISOS Procedures," *Nature Energy* 5, no. 1 (2020): 35–49, https://doi.org/10.1038/s41560-019-0529-5.

28. K. T. Tanko, Z. Tian, S. Raga, H. Xie, E. A. Katz, and M. Lira-Cantu, "Stability and Reliability of Perovskite Photovoltaics: Are We There Yet?," *MRS Bulletin* 50, no. 4 (2025): 512–525, https://doi.org/10.1557/s43577-025-00863-5.

29. M. I. Hossain and B. Aïssa, "Effect of Structure, Temperature, and Metal Work Function on Performance of Organometallic Perovskite Solar Cells," *Journal of Electronic Materials* 46, no. 3 (2017): 1806–1810, https://doi.org/10.1007/s11664-016-5232-8.

30. X. Rong, H. Tian, W. Bi, et al., "Impact of Metal Electrode Work Function of $CH_3NH_3PbI_3$/p-Si Planar Heterojunction Perovskite Solar Cells," *Solar Energy* 158 (2017): 424–431, https://doi.org/10.1016/j.solener.2017.08.050.

31. F. Jiang, Y. Shi, T. R. Rana, et al., "Improved Reverse Bias Stability in p–i–n Perovskite Solar Cells With Optimized Hole Transport Materials and Less Reactive Electrodes," *Nature Energy* 9, no. 10 (2024): 1275–1284, https://doi.org/10.1038/s41560-024-01600-z.

32. C. F. Glover, J. McGettrick, G. Williams, T. M. Watson, and D. Bryant, "A Scanning Kelvin Probe Investigation of the Interaction of PEDOT: PSS Films With Metal Surfaces And Potential Corrosion Protection Properties," *Journal of The Electrochemical Society* 162, no. 10 (2015): H799–H805, https://doi.org/10.1149/2.0761510jes.

33. Y. J. Hofstetter, I. García-Benito, F. Paulus, S. Orlandi, G. Grancini, and Y. Vaynzof, "Vacuum-Induced Degradation of 2D Perovskites," *Frontiers in Chemistry* 8 (2020): 66, https://doi.org/10.3389/fchem.2020.00066.

34. M. De Keersmaecker, P. Dietrich, M. Bahri, N. D. Browning, N. R. Armstrong, and E. L. Ratcliff, "Activated Corrosion and Recovery in Lead Mixed-Halide Perovskites Revealed by Dynamic Near-Ambient Pressure X-ray Photoelectron Spectroscopy," *Journal of the American Chemical Society* 147, no. 10 (2025): 8881–8892, https://doi.org/10.1021/jacs.5c00668.

35. T. Li, C. Ma, Y. Li, et al., "In Situ Preparation of Chromium Carbide–Modified Carbon Nanofibers as Functional Electrocatalyst for Polysulfide Reduction in Lithium/Sulfur Batteries," *Ionics* 28, no. 4 (2022): 1701–1711, https://doi.org/10.1007/s11581-021-04431-0.

36. Y. Li, Y. Gao, B. Xiao, et al., "The Electronic, Mechanical Properties and Theoretical Hardness of Chromium Carbides by First-Principles Calculations," *Journal of Alloys and Compounds* 509, no. 17 (2011): 5242–5249, https://doi.org/10.1016/j.jallcom.2011.02.009.

37. A. Zikin, I. Hussainova, C. Katsich, E. Badisch, and C. Tomastik, "Advanced Chromium Carbide-Based Hardfacings," *Surface and Coatings Technology* 206, no. 19 (2012): 4270–4278, https://doi.org/10.1016/j.surfcoat.2012.04.039.

38. B. Bozzini, G. P. De Gaudenzi, M. Serra, A. Fanigliulo, and F. Bogani, "Corrosion Behaviour of WC-Co Based Hardmetal In Neutral Chloride And Acid Sulphate Media," *Materials and Corrosion* 53, no. 5 (2002): 328–334, https://doi.org/10.1002/1521-4176(200205)53:5⟨328.

39. L. Sun, X. Ji, L. Zhao, et al., "First Principles Investigation of Binary Chromium Carbides Cr_7C_3, Cr_3C_2 and $Cr_{23}C_6$: Electronic Structures, Mechanical Properties and Thermodynamic Properties Under Pressure," *Materials* 15, no. 2 (2022): 558, https://doi.org/10.3390/ma15020558.

40. Q. Cao, S. Liu, and D. Li, "Theoretical Studies for Stability, Mechanical Properties, Electronic Properties and Debye Temperature of Novel Cr_2C Structures," *Journal of Materials Science* 57, no. 40 (2022): 18969–18979, https://doi.org/10.1007/s10853-022-07802-8.

41. T. Ohmi, A. Ohki, M. Nakamura, et al., "The Technology of Chromium Oxide Passivation on Stainless Steel Surface," *Journal of the Electrochemical Society* 140, no. 6 (1993): 1691–1699, https://doi.org/10.1149/1.2221625.

42. M. Koudelkova, J. Augustynski, and H. Berthou, "On the Composition of the Passivating Films Formed on Aluminum in Chromate Solutions," *Journal of the Electrochemical Society* 124, no. 8 (1977): 1165–1168, https://doi.org/10.1149/1.2133520.

43. J. B. Wagner, K. R. Lawless, and A. T. Gwathmey, "Passivation of Aluminum by Chromate Solutions," *Journal of the Electrochemical Society* 114, no. 10 (1967): 1001–1007.

44. T. Niewelt, W. Kwapil, M. Selinger, A. Richter, and M. C. Schubert, "Long-Term Stability of Aluminum Oxide Based Surface Passivation Schemes Under Illumination at Elevated Temperatures," *IEEE Journal of Photovoltaics* 7, no. 5 (2017): 1197–1202, https://doi.org/10.1109/JPHOTOV.2017.2713411.

45. R. Kotipalli, R. Delamare, O. Poncelet, X. Tang, L. A. Francis, and D. Flandre, "Passivation Effects of Atomic-Layer-Deposited Aluminum Oxide," *EPJ Photovoltaics* 4 (2013): 45107, https://doi.org/10.1051/epjpv/2013023.

46. T. Xu, Y. Xu, N. Liao, Y. Li, and M. Nath, "High-Temperature Chemical Stability of Cr(III) Oxide Refractories in the Presence of Calcium Aluminate Cement," *Materials* 14, no. 21 (2021): 6590, https://doi.org/10.3390/ma14216590.

47. J. Ramm, M. Ante, H. Brändle, A. Neels, A. Dommann, and M. Döbeli, "Thermal Stability of Thin Film Corundum-Type Solid Solutions of $(Al_{1-x}Cr_x)_2O_3$ Synthesized Under Low-Temperature Non-Equilibrium Conditions," *Advanced Engineering Materials* 9, no. 7 (2007): 604–608, https://doi.org/10.1002/adem.200700081.

48. D. Gozzi, P. L. Cignini, E. Stampa, B. Alfonsi, and M. Petrucci, "Some Evidences on the Instability of Cr-Cr_2O_3 Electrode in the Electrochemical Oxygen Sensors At Steelmaking Temperatures," *Transactions of the Iron and Steel Institute of Japan* 23, no. 3 (1983): 205–212, https://doi.org/10.2355/isijinternational1966.23.205.

49. O. M. Cintho, E. A. P. Favilla, and J. D. T. Capocchi, "Mechanical–Thermal Synthesis of Chromium Carbides," *Journal of Alloys and Compounds* 439, no. 1 (2007): 189–195, https://doi.org/10.1016/j.jallcom.2006.03.102.

50. S. S. Chatha, H. S. Sidhu, and B. S. Sidhu, "Characterisation and Corrosion-Erosion Behaviour of Carbide-based Thermal Spray Coatings," *Journal of Minerals and Materials Characterization and Engineering* 11, no. 06 (2012): 569–586, https://doi.org/10.4236/jmmce.2012.116041.

51. L. Zhou, J. Su, Z. Lin, et al., "Theoretical and Experimental Investigation of Mixed Pb–In Halide Perovskites," *The Journal of Physical Chemistry C* 122, no. 28 (2018): 15945–15953, https://doi.org/10.1021/acs.jpcc.8b05267.

52. S. S. Mammana, A. Greatti, F. H. Luiz, et al., "Study of Wet Etching Thin Films of Indium Tin Oxide in Oxalic Acid by Monitoring the Resistance," *Thin Solid Films* 567 (2014): 20–31, https://doi.org/10.1016/j.tsf.2014.07.027.

53. R. A. Kerner and B. P. Rand, "Electrochemical and Thermal Etching of Indium Tin Oxide by Solid-State Hybrid Organic–Inorganic Perovskites," *ACS Applied Energy Materials* 2, no. 8 (2019): 6097–6101, https://doi.org/10.1021/acsaem.9b01356.

54. A. Senocrate, T.-Y. Yang, G. Gregori, G. Y. Kim, M. Grätzel, and J. Maier, "Charge Carrier Chemistry in Methylammonium Lead Iodide," *Solid State Ionics* 321 (2018): 69–74, https://doi.org/10.1016/j.ssi.2018.03.029.

55. S.-M. Bang, S. S. Shin, N. J. Jeon, et al., "Defect-Tolerant Sodium-Based Dopant in Charge Transport Layers for Highly Efficient and Stable

Perovskite Solar Cells," *ACS Energy Letters* 5, no. 4 (2020): 1198–1205, https://doi.org/10.1021/acsenergylett.0c00514.

56. N. Li, S. Tao, Y. Chen, et al., "Cation and Anion Immobilization Through Chemical Bonding Enhancement With Fluorides for Stable Halide Perovskite Solar Cells," *Nature Energy* 4, no. 5 (2019): 408–415, https://doi.org/10.1038/s41560-019-0382-6.

57. W. Zhao, Z. Yao, F. Yu, and D. Yang, "Alkali Metal Doping for Improved $CH_3NH_3PbI_3$ Perovskite Solar Cells," *Advanced Science* 5, no. 2 (2018): 1700131, https://doi.org/10.1002/advs.201700131.

58. R. M. Tichane and G. B. Carrier, "Microstructure of a Soda-Lime Glass Surface," *Journal of the American Ceramic Society* 44, no. 12 (1961): 606–610, https://doi.org/10.1111/j.1151-2916.1961.tb11667.x.

59. L. Lanzetta, L. Huerta Hernandez, B. Vishal, et al., "Tin–Lead Perovskite Solar Cells with Enhanced Reverse Bias Stability," *ACS Energy Letters* 10, no. 5 (2025): 2093–2095, https://doi.org/10.1021/acsenergylett.5c00727.

60. R. A. Z. Razera, D. A. Jacobs, F. Fu, et al., "Instability of p–i–n Perovskite Solar Cells Under Reverse Bias," *Journal of Materials Chemistry A* 8, no. 1 (2019): 242–250, https://doi.org/10.1039/C9TA12032G.

61. K. Yan, M. Peng, X. Yu, et al., "High-Performance Perovskite Memristor Based On Methyl Ammonium Lead Halides," *Journal of Materials Chemistry C* 4, no. 7 (2016): 1375–1381, https://doi.org/10.1039/C6TC00141F.

62. C. Wang, L. Huang, Y. Zhou, et al., "Perovskite Solar Cells in the Shadow: Understanding the Mechanism of Reverse-Bias Behavior Toward Suppressed Reverse-Bias Breakdown and Reverse-Bias Induced Degradation," *Advanced Energy Materials* 13, no. 9 (2023): 2203596, https://doi.org/10.1002/aenm.202203596.

63. D. Bogachuk, K. Saddedine, D. Martineau, et al., "Perovskite Photovoltaic Devices With Carbon-Based Electrodes Withstanding Reverse-Bias Voltages up to −9 V and Surpassing IEC 61215:2016 International Standard," *Solar RRL* 6, no. 3 (2022): 2100527, https://doi.org/10.1002/solr.202100527.

64. S. Hu, J. Wang, P. Zhao, et al., "Steering Perovskite Precursor Solutions for Multijunction Photovoltaics," *Nature* 639 (2024): 93–101, https://doi.org/10.1038/s41586-024-08546-y.

65. S. Liu, J. Li, W. Xiao, et al., "Buried Interface Molecular Hybrid for Inverted Perovskite Solar Cells," *Nature* 632, no. 8025 (2024): 536–542, https://doi.org/10.1038/s41586-024-07723-3.

66. Z. Wang, L. Zeng, T. Zhu, et al., "Suppressed Phase Segregation for Triple-Junction Perovskite Solar Cells," *Nature* 618, no. 7963 (2023): 74–79, https://doi.org/10.1038/s41586-023-06006-7.

67. S. Hu, J. Pascual, W. Liu, et al., "A Universal Surface Treatment for p–i–n Perovskite Solar Cells," *ACS Applied Materials & Interfaces* 14, no. 50 (2022): 56290–56297, https://doi.org/10.1021/acsami.2c15989.

68. S. Hu, K. Otsuka, R. Murdey, et al., "Optimized Carrier Extraction at Interfaces for 23.6% Efficient Tin–Lead Perovskite Solar Cells," *Energy & Environmental Science* 15, no. 5 (2022): 2096–2107, https://doi.org/10.1039/D2EE00288D.

69. J. C. de Mello, H. F. Wittmann, and R. H. Friend, "An Improved Experimental Determination Of External Photoluminescence Quantum Efficiency," *Advanced Materials* 9, no. 3 (1997): 230–232, https://doi.org/10.1002/adma.19970090308.

70. T. Kirchartz, J. A. Márquez, M. Stolterfoht, and T. Unold, "Photoluminescence-Based Characterization of Halide Perovskites for Photovoltaics," *Advanced Energy Materials* 10, no. 26 (2020): 1904134, https://doi.org/10.1002/aenm.201904134.

71. M. Taddei, S. Jariwala, R. J. E. Westbrook, et al., "Interpreting Halide Perovskite Semiconductor Photoluminescence Kinetics," *ACS Energy Letters* 9, no. 6 (2024): 2508–2516, https://doi.org/10.1021/acsenergylett.4c00614.

72. National Aeronautics and Space Administration (NASA), "TPSX Material Properties Database" tpsx.arc.nasa.gov/MaterialsDatabase (2024).

73. D. R. Lide, G. Baysinger, L. I. Berger, et al., *CRC Handbook of Chemistry and Physics* (CRC Press, 1995).

74. A. Weatherford, Geological Survey Mineral Industry Surveys Gold https://www.usgs.gov/centers/national-minerals-information-center/gold-statistics-and-information (2024).

75. Mineral Industry Surveys U.S. Geological Survey, www.usgs.gov/centers/national-minerals-information-center/mineral-industry-surveys (2024).

76. L. P. Bloomberg, "Precious and Industrial Metals" (2024), www.bloomberg.com/markets/commodities/futures/metals.

77. A. V. Naumkin, A. Kraut-Vass, S. W. Gaarenstroom, and C. J. Powell, NIST X-ray Photoelectron Spectroscopy Database, NIST Standard Reference Database Number 20, Version 4.1 (Gaithersburg, MD: National Institute of Standards and Technology, 2012), https://doi.org/10.18434/T4T88K.

78. A. Y. Lee, C. J. Powell, J. M. Gorham, A. Morey, J. H. J. Scott, and R. J. Hanisch, "Development of the NIST X-Ray Photoelectron Spectroscopy (XPS) Database, Version 5," *Data Science Journal* 23, no. 1 (2024): 45, https://doi.org/10.5334/dsj-2024-045.

79. J. D. McGettrick, E. Speller, Z. Li, W. C. Tsoi, J. R. Durrant, and T. Watson, "Use of Gas Cluster Ion Source Depth Profiling To Study The Oxidation Of Fullerene Thin Films by XPS," *Organic Electronics* 49 (2017): 85–93, https://doi.org/10.1016/j.orgel.2017.06.022.

80. M. C. Biesinger, B. P. Payne, A. P. Grosvenor, L. W. Lau, A. R. Gerson, and R. S. C. Smart, "Resolving surface chemical states in XPS analysis of first row transition metals, oxides and hydroxides: Cr, Mn, Fe, Co and Ni," *Applied Surface Science* 257, no. 7 (2011): 2717–2730, https://doi.org/10.1016/j.apsusc.2010.10.051.

Supporting Information

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